STANDARD OPERATING PROCEDURE FOR THE VOCs DATA QUALITY CONTROL & VALIDATION OF THE AGILENT 7890B and 8890 AUTOMATED GAS CHROMATOGRAPH SYSTEM AND ASSOCIATED COMPONENTS.

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1.0 PURPOSE

This Standard Operating Procedure (SOP) delineates the steps required to verify and validate the data associated with the Markes CIA Advantage-xr, Kori-xr, UNITY-xr, and Agilent 7890b and 8890 automated gas chromatograph (AutoGC) systems and their associated components. By following this SOP, the data validator ensures that through data verification and validation, the AutoGC systems produce the appropriate quantity and quality of data required to meet data quality objectives (DQOs) set forth in the *Photochemical Assessment Monitoring Station (PAMS) Network Quality Assurance Project Plan (QAPP) for Utah Air Monitoring.*

2.0 SCOPE AND APPLICABILITY

2.1 Intent for Use

This procedure is intended for use by the Air Monitoring Section's AutoGC site validators. It is assumed that an AutoGC validator has a technical background with knowledge of air pollution, meteorology, data analysis techniques, federal and state environmental law, and monitoring programs within the DAQ. It is also assumed that an AutoGC site validator has been trained to have a fundamental understanding of the data collected by the GC and GC component systems, how the data collection is controlled by the site computer's software, and how the data is analyzed using the EZChrom software and ORSAT Merlin AutoGC Xplorer (MAX) cloud-based data management software. and has the required user accounts, permissions, and tools needed to perform the functions outlined in this SOP.

2.2 Valid Time frame

The SOP covers the timeframe between the effective date on the original copy until superseded by another approved SOP. Previous versions of the original hardcopy SOP are archived for future reference.

- This procedure is reviewed annually and revised every five years at a minimum.

3.0 METHOD SUMMARY

On an hourly cycle, the Markes CIA Advantage-xr, Kori-xr, UNITY-xr, and Agilent 7890b (system using Helium as the carrier gas) or Agilent 8890 (system using Hydrogen as the carrier gas) AutoGC system draws in ambient air from the inlet probe to collect and preconcentrate VOCs from the sampled atmosphere and subsequently separate the VOCs for detection via a flame ionization detector (FID). A new sample commences at the top of each hour.

Within the GC, the compounds pass through the Boiling Point (BP) separation column and are routed through a Dean's switch, which directs the lighter hydrocarbons (C2-C6) through a subsequent alumina Porous Layer Open Tubular (PLOT) separation column. The PLOT column separates the lighter hydrocarbons by weight as they pass through before being ignited by a hydrogen-fueled flame ionization detector (FID). Compounds generally elute from lighter to heavier by weight, but many factors such as flow, humidity, and temperature can affect the elution of PLOT column compounds. The FID detects ions produced by the ignition of compounds and translates them to an electronic signal. Following the elution of 1-hexene, the Dean's switch flips and heavier compounds (C6-C12) pass solely through the BP column for separation. These heavier compounds elute to a separate FID from lower to higher boiling points which is directly correlated to lighter to heavier weights respectively.

Compounds are identified by their retention time in the analytical method within the Agilent EZChrom software. The amperage responses of the FIDs are converted to a concentration in parts per billion carbon (ppbC) based on the response factor determined during calibration. The response factor is based on the FID response of propane or butane (C2-C6) or benzene or toluene (C6-C12), depending on the calibrant selected for each column, in a multipoint calibration. Data is then stored on the site computer, transmitted to the ORSAT MAX website, and archived manually to the U:\ network drive at U:\PLAN\AMC\GC-Agilent+Markes\Data

4.0 PROCEDURE

4.1 Background

This section describes in detail the procedures for verifying and validating the data generated by DAQ's hourly Auto-GC analyzers.

Auto-GCs are scheduled to operate 24 hours a day throughout the year with the majority of hours reserved for ambient sampling. Other hours are reserved for quality control runs or are interrupted by instrument maintenance. The

instruments are designed to identify target volatile organic compounds (VOCs) that are of interest primarily because of their contribution to ground-level ozone formation and/or toxicity. Appendix A, the AutoGC Master Compound List, includes current target VOCs as required by the DAQ PAMS QAPP.

The AutoGC data validator verifies that the target compounds are identified, quantified, and reported according to the DAQ PAMS QAPP DQOs. Validators and operators must communicate with each other to maximize data quality and data capture. AutoGC site operators, network administrators, and data validators are responsible for ensuring the retrieval and storage of data collected from the remote sites.

4.2 Routine Operations

4.2.1 Daily Tasks

- Log in to ORSAT MAX website and check on file upload status. Missing files or the upload status not being current could indicate a problem with communications or the AutoGC.
- On ORSAT MAX, check daily CVS recoveries and Blank results.
- Communicate with data operators if problems arise.

4.2.2 Weekly Tasks

- Log in to ORSAT MAX, check weekly precision, RTS and LCS recoveries results.
- On ORSAT MAX, review the weekly data trends. Communicate with data operators if problems are detected.

4.2.3 Monthly Tasks

- Conduct review of measurements, reprocess files as necessary.
- Document review and reprocess in Monthly Data Validation Report (MDVR).
- Upload reprocessed files into ORSAT MAX.
- Lock ORSAT MAX upon the completion of the review and create AQS upload files.
- Submit AQS files to Uploader.
- Inform data operators of any consistent issues with the measurements.
- Archive and zip final documentation (logbook, MDVR, calibration report, etc.)

4.2.4 Methods

The Agilent GC analytical method is where most data analysis occurs. The EZChrom software processes the chromatography data, calculates the picoamperage values, integrates peaks, calculates peak areas, converts peak areas to concentration values, and assigns compound names to peaks.

4.3 Data Review

After a days' worth of data is collected from the analyzer, a program on the site computer will archive that day's data files into a Daily-ZIP file. A Daily-ZIP file could include more than one day's worth of data if there were communications failures, power failures, maintenance, or other aberrant site conditions. At least once a month, a member of the validation team will manually copy the Daily-ZIP file from the site computer to U:\PLAN\AMC\GC-Agilent+Markes\Data\ORIGINAL .

A program on the site computer will automatically transfer the day's data to ORSAT MAX.

Use 7-Zip, WinZip, or its equivalent to open newly transferred Daily-ZIP files and extract all of the files except.tx1 files to your personal computer in a place such as C:\Users\[*username]*\Documents\AutoGC Data\[*Site ID]*\[*YYYY*]\[*YYYYMM*]

the following applicable QC reports as prescribed, or whenever a QC failure occurs, following the steps outlined below. Look for general trends such as compound percent recoveries declining or increasing over time.

True QC failures will also affect the quality of ambient data. For instance, a CVS or Blank may fail due to a trap failure. The trap failing will not only affect the QC data, but it will also affect the quality of ambient data. True QC failures will result in the invalidation of ambient data.

Sometimes an event can be identified that causes subsequent QCs to fail (Type I Causative Event). In these cases, data prior to the causative event is not compromised and is valid, but data after the causative event is invalid. Examples of a Type I causative event include: air conditioning failures, sample pump failures, multipoint calibrations, and the flame ionization detector being extinguished.

QC failures can also be caused by events that do not compromise the integrity of the ambient data before and after the QC (Type II Causative Event). For instance, low QC canister pressure may result in a CVS failure, but ambient data will not be affected. Other examples include: power outages that interrupt the QC sample collection, valves not switching to allow QC gas to be collected, electronic spiking, target compound misidentification, peak splitting/coelution issues, or a bad QC canister.

4.3.1 Calibration Verification Standard (CVS)- Description and Reports

The CVS is a daily analysis of a 1 ppmV standard comprised of 14 components blended at the site with humidified zero air. A list of compounds present in the CVS are shown in **Table 1**.

Table 1. List of the compounds included in the CVS and LCS standards.

PLOT Column ***(Listed in elution order)***	**Boiling Point (BP) Column** ***(Listed in elution order)***
Ethane	n-Hexane
Propane	Benzene
n-Butane	Toluene
Acetylene	m&p-Xylene
n-Pentane	n-Propylbenzene
1,3-Butadiene	1,2,4-Trimethylbenzene
2-Methylpentane	p-Diethylbenzene

Under normal system operations, the CVS is scheduled to run daily and is followed by a method blank. Instrument drift is considered acceptable when the percent recoveries for the compounds selected as a calibrant in each column compared to canister specific concentrations fall within **70-130** percent. If the recoveries for the compound selected as calibrant (propane or butane for the PLOT and benzene or toluene for the BP) for each column are outside of this range, an entire column failure procedure applies. For the CVS only, a partial column failure procedure applies when the percent recoveries for any other calibrant fall outside 70-130 percent.

To see the recovery results of all the calibrants in the CVS analyses over a given time range at a particular site:

- From ORSAT MAX home, select the relevant AutoGC monitoring site from the map.
- At the top right of the monitoring site's homepage, select the "Generate Charts" link.

- Under “Choose Chart Report” select “Standard Recoveries and Amount”.
- Select the relevant date range, the radio button next to “Recovery”, the relevant monitoring site, the checkbox next to “CVS”, the radio button next to “All Components Channel A and B”, and the radio button next to “No” under “Combine Channel A and B Chart”, then click the “Generate Chart” button.
- There are two ways to review the data. The time series graph will be prominent, and the “Show Data Table” button below any graph can be selected to view the raw recoveries in a data table.

Refer to **Table 2** for Response Actions when a CVS does not meet the listed Acceptance Criteria. Notify the operators and other personnel for all QC failures as necessary to determine if further investigation is warranted.

Table 2. PAMS Data Quality Objectives for the CVS used in the DAQ AutoGC program, including response actions for tests that are out of limits.

QC Check	Purpose	Frequency	Acceptance Criteria	Response Action to Failures
Calibration Verification Standard (CVS)	To assess instrument drift and to ensure continued instrument calibration and instrument integrity	Daily-nightly, preferable between 22:00 and 3:00	• Calibrant species percent recoveries within 70-130 % • Acetylene percent recovery within 70-130% • Assuming calibrant species pass and ignoring acetylene, all other calibrants' percent recoveries within 70-130%	• Investigate chromatogram for retention time shifts which may result in peak misidentification. Investigate for instrument contamination resulting in co-eluting peaks. Investigate for system leaks or trap malfunction resulting in low recovery. Unless technical justification is provided to explain nonconformance, qualify as "QX" in AQS all samples for affected compounds since the most recent passing CCV considering: • If there is a Type I Causative Event, flag from the causative event forward to the next passing CVS. • If there is a Type II Causative Event, flag only that CVS hour. • If there are neither Type I nor Type II Causative Events and calibrant for the column (s) fails, flag all PLOT or and BP column data backward and forward to passing CVSs. TNMHC and TNMTC should be automatically flagged when the "Column(s)" is invalidated. • If there are neither Type I nor Type II Causative Events and acetylene fails, flag acetylene backward and forward to acetylene-passing CVSs. • If there are neither Type I nor Type II Causative Events and (ignoring acetylene) a calibrant fails, flag the calibrant and every target compound after the previous passing calibrant (by elution order) up to but not including the next passing calibrant (by elution order) backward and forward to calibrant-passing CVSs.

Calibrant species will be propane or butane for the Plot column and benzene or toluene for the BP column

4.3.2 Precision Check – Description and Reports

A precision check consisting of two back-to-back CVS samples is performed once a week to determine analytical reproducibility. A precision check is considered acceptable when the relative percent difference (RPD) between the two CVS runs is less than **30**% for propane or butane and benzene or Toluene, depending on which is the calibrant for each column

Refer to **Table 3** for Response Actions when a Precision Check does not meet the listed Acceptance Criteria. Notify the operators and other personnel for all QC failures as necessary to determine if further investigation is warranted.

Table 3. PAMS Data Quality Objectives for the Precision Check used in the DAQ AutoGC program, including response actions for tests that are out of limits.

QC Check	Purpose	Frequency	Acceptance Criteria	Response Action to Failures
Precision Check (Two consecutive CVS runs)	To assess analytical precision	Weekly	• Calibrant species RPD <30%	• Investigate system for carryover, contamination, leaks, or suppression, as indicated by trends in compound behavior. Qualify ambient sample data for affected compounds since the last passing precision check as "QX" in AQS considering: • If there is a Type II Causative Event and the compound selected as calibrant in each column fails the weekly Precision Check, then the data are valid. Flag only the CVS hour(s). • If there is neither a Type I nor Type II Causative Event and the compound selected as calibrant for each column fails the weekly Precision Check, check the day-to-day precision of the failed compound backward and forward a week. • If there is a Type I Causative Event and the compound selected as calibrant for each column fails the weekly Precision Check, check the day-to-day precision of the failed compound forward from the causative event to the next passing precision check. • If the day-to-day precision passes for the compound(s) that failed the weekly Precision Check, the data are valid. • If the day-to-day precision fails one day or more for the compound(s) that failed the weekly Precision Check, flag all data for that column backward and forward to valid weekly Precision Checks. TNMHC and TNMTC are automatically invalidated when "Column A" or "Column B" are invalidated.

Calibrant species will be propane or butane for the Plot column and toluene for the BP column

4.3.3 Method Blank – Description and Reports

A method blank is a humidified air matrix, void of contaminants, processed under the same conditions as samples. Results are assessed to determine if there is carry-over and contamination residing in the system. The daily blank is performed immediately after the daily CVS and it is necessary to check for system cleanliness. Other blanks will run after multipoint calibrations and other system maintenance. The blank is considered acceptable when individual concentrations of target compounds are less than MDL or 0.5 ppbC and Total Non-Methane Hydrocarbons (TNMHC) and Total Non-Methane Target Compounds (TNMTC) are less than 10.0 ppbC.

To see the results of Method Blanks over a given time range at a particular site:

- From ORSAT MAX home, select the relevant AutoGC monitoring site from the map.
- At the top right of the monitoring site's homepage, select the "Generate Charts" link.
- Under "Choose Chart Report" select "Standard Recoveries and Amount".
- Select the relevant date range, the radio button next to "Amount", the relevant monitoring site, the checkbox next to "Blank", the checkboxes next to both "TNMTC" and "TNMHC", the radio button next to "All Components Channel A and B", and the radio button next to "No" under "Combine Channel A and B Chart", then click the "Generate Chart" button.
- There are two ways to review the data. The time series graph will be prominent, and the "Show Data Table" button below any graph can be selected to view the raw amounts in a data table.

Refer to **Table 4** for Response Actions when a Method Blank does not meet the listed Acceptance Criteria. Notify the operators and other personnel for all QC failures as necessary to determine if further investigation is warranted.

Table 4. PAMS Data Quality Objectives for the Method Blank test used in the DAQ AutoGC program, including response actions for tests that are out of limits.

QC Check	Purpose	Frequency	Acceptance Criteria	Response Action to Failures
Method (Analytical) Blank	System Contribution	Daily	• All target compounds ≤ the determined MDL or 0.5 ppbC, whichever is lower • TNMHC <10.0 ppbC	• Analyze another blank, if possible, to investigate potential carryover from a high concentration sample. Investigate system for contamination. Unless technical justification is provided to explain nonconformance, qualify as "LB" in AQS all samples for affected compounds since the last passing SB considering: • If a blank failure occurs following a Type I[a] Causative Event, flag the appropriate parameters from the causative event to the next passing blank. • If there is a Type II Causative Event, flag only the Blank hour. • If there is neither a Type I nor Type II Causative Event and a target compound above the MDL will be qualified with the QX flag or nullified with the AS flag. The flag will apply to that compound backward and forward to passing blanks. • If there is neither a Type I nor Type II Causative Event and the TNMHC is greater than or equal to 10 ppbC, flag TNMHC and Total Non-Methane Target Compounds (TNMTC) backward and forward to passing blanks.

4.3.4 Laboratory Calibration Standard (LCS) – Description and Reports

The LCS is a 5 ppbV second source standard control sample that is scheduled every week to assess calibration accuracy. The LCS, like the CVS, is comprised of 14 components as shown in **Table 1**. The LCS sample is introduced directly into the instrument without blending with humidified zero air. The LCS can be used as an alternative to the CVS for calibration verification purposes if the CVS is faulty due to problems not related to the instrument, such as a bad CVS canister, or for individual compounds if the LCS can prove that the CVS is not a reliable measure of percent recovery (such as in the case of acetylene which can tend to have a lower recovery

in the CVS). Since the LCS sample is injected directly into the instrument without passing through the blender, the resulting LCS recoveries can indicate if the calibration system and AutoGC are both drifting. Instrument drift is considered acceptable when the percent recoveries for the compounds selected as calibrants for each of the columns falls within **70-130** percent.

To see the recovery results of all the calibrants in the LCS analyses over a given time range at a particular site:

- From ORSAT MAX home, select the relevant AutoGC monitoring site from the map.
- At the top right of the monitoring site's homepage, select the "Generate Charts" link.
- Under "Choose Chart Report" select "Standard Recoveries and Amount".
- Select the relevant date range, the radio button next to "Recovery", the relevant monitoring site, the checkbox next to "LCS", the radio button next to "All Components Channel A and B", and the radio button next to "No" under "Combine Channel A and B Chart", then click the "Generate Chart" button.
- There are two ways to review the data. The time series graph will be prominent, and the "Show Data Table" button below any graph can be selected to view the raw recoveries in a data table.

Refer to **Table 5** for Response Actions when an LCS does not meet the listed Acceptance Criteria. Notify the operators and other personnel for all QC failures as necessary to determine if further investigation is warranted. **Note:** Sometimes low-pressure canisters or incorrect canister preparation lead to recoveries to be outside of the desired value range

Table 5. PAMS Data Quality Objectives for the LCS used in the DAQ AutoGC program, including response actions for tests that are out of limits.

QC Check	Purpose	Frequency	Acceptance Criteria	Response Action to Failures
Laboratory Calibration Standard (LCS)	Second-source standard to assess calibration accuracy	Weekly	• Calibrant species percent recoveries within 70-130%	• Investigate the discrepancy between CVS and LCS. Investigate chromatogram for retention time shifts which may result in peak misidentification. Investigate for instrument contamination resulting in co-eluting peaks. Investigate for system leaks or trap malfunction resulting in low recovery. Unless technical justification is provided to explain nonconformance, minimally qualify as "QX" and potentially invalidate as "AS" samples for affected compounds since the last acceptable LCS considering: • If there is a Type II Causative Event, flag only the LCS hour. • If there is not a Type II Causative Event, review Calibrant species and determine cause of failure to determine if instrument drift has occurred. Request the operator to run a multipoint calibration • If there is neither a Type I nor Type II Causative Event and the operator makes an adjustment prior to running a multipoint calibration, invalidate PLOT Column (if propane or butane, whichever is the calibrant, fails) and/or BP Column (if benzene or toluene, whichever is the calibrant, fails) from the last passing LCS to the adjustment. • If there is neither a Type I nor Type II Causative Event, the operator does not make an adjustment prior to the multipoint calibration, and the new response factor failed (RF >10%), for any of the selected calibrants invalidate the respective column from the last passing LCS to the multipoint calibration. • If there is a Type I Causative Event and the operator makes an adjustment prior to the multipoint calibration, invalidate all the compounds in the column for which the calibrant failed from the causative event to the adjustment.

				• If there is a Type I Causative Event and the operator does not make an adjustment prior to the multipoint calibration, apply the response factor of the multipoint calibration for the appropriate column to the causative event. Check to see if the LCS still fails. If it doesn't, no action is needed. If it still fails, invalidate the appropriate column from the causative event until a passing LCS is achieved.

4.3.5 Retention Time Standard (RTS) – Description and Reports

A RTS sample is scheduled every week and after any maintenance period to verify compound identification. This 57-component standard comprises all of the PAMS Program target compounds. It is analyzed to help assess elution order and retention time shifts in the chromatographic method.

To see the results for all compounds in the RTS analyses over a given time range at a particular site:

- From ORSAT MAX home, select the relevant AutoGC monitoring site from the map.
- At the top right of the monitoring site's homepage, select the "Generate Charts" link.
- Under "Choose Chart Report" select "Standard Recoveries and Amount".
- Select the relevant date range, the radio button next to "Amount", the relevant monitoring site, the checkbox next to "RTS", the radio button next to "All Components Channel A and B", and the radio button next to "No" under "Combine Channel A and B Chart", then click the "Generate Chart" button.
- There are two ways to review the data. The time series graph will be prominent, and the "Show Data Table" button below any graph can be selected to view the raw amounts in a data table.

Refer to Response Actions when an RTS does not meet the listed Acceptance Criteria. Notify the operators and other personnel for all QC

failures as necessary to determine if further investigation is warranted. Compounds included in the RTS standard are shown in **Table 6**

Table 6. List of the compounds included in the RTS standard

PLOT column	PLOT column (cont.)	BP column	BP column (cont.)
Ethane	3-methylpentane	N-hexane	Styrene
Ethylene	Isoprene	Methylcyclopentane	O-xylene
Propane	2-methyl-1-pentene	2,4-dimethylpentane	N-nonane
Propylene	1-hexene	Benzene	Iso-propylbenzene
Iso-butane		Cyclohexane	N-propylbenzene
N-butane		2-methylhexane	M-ethyltoluene
Acetylene		2,3-dimethylpentane	P-ethyltoluene
Trans-2-butene		3-methylhexane	1,3,5-tri-m-benzene
1-butene		2,2,4-trimethylpentane	O-ethyltoluene
Cis-2-butene		N-heptane	1,2,4-tri-m-benzene
Cyclopentane		Methylcyclohexane	N-decane
Iso-pentane		2,3,4-trimethylpentane	1,2,3-tri-m-benzene
N-pentane		Toluene	M-diethylbenzene
1,3-butadiene		2-methylheptane	P-diethylbenzene
Trans-2-pentene		3-methylheptane	N-undecane
1-pentene		N-octane	N-dodecane
Cis-2-pentene		Ethylbenzene	
2,2-dimethylbutane		M&P-xylene	

Table 7. PAMS Data Quality Objectives for the RTS used in the DAQ AutoGC program, including response actions for tests that are out of limits.

QC Check	Purpose	Frequency	Acceptance Criteria	Response Action
Retention Time Standard (RTS) (also called "57-Component Standard")	To qualitatively assess retention time shifts and processing methods	Weekly	• Target compounds are integrated and identified correctly	• Review the previous week's ambient and QC check sample data to evaluate events resulting in retention time shift. May require reassignment or adjustment of retention time windows and reprocessing of data collected since the most recent CCV or RTS. Unless technical justification is provided to explain nonconformance, associated ambient sample data will be invalidated as "BH" for compounds whose identities cannot be confirmed. Considering: • If there is either a Type I or Type II Causative Event that affects the entire RTS, invalidate the RTS hour. • If there is either a Type I or Type II Causative Event for the target compounds not being integrated and the same compounds are integrated correctly in the surrounding ambient air data, then no action is necessary. • If there is neither a Type I nor Type II Causative Event for the target compound misidentification/lack of integration, the validator may do one or more of the following: ○ Contact the operator to determine the cause and proceed according to the findings. ○ Request the operator to optimize the current method ○ Reprocess (or ask the operator to reprocess) affected data as necessary

4.3.6 Annual Performance Audit– Description and Reports

An annual performance audit is performed in accordance with the PAMS QAPP. The audit sample consists of a selection of target compounds whose identities and concentrations are initially only known to the contractor preparing the audit canister and the auditor. The audit sample is introduced directly into the instrument without blending. The results of the audit sample are then compared with the expected values. Results of the audit sample are considered acceptable when recoveries fall within **70-130** percent.

The audit results will initially only be visible to the auditor. After reviewing the data, the auditor will provide a report to the site operator, subject matter expert. If an audit fails, data will not necessarily be invalidated: the auditor will consult with the site operator and data validator on the next steps.

4.3.7 Multipoint Calibration – Description and Validation Acceptance Criteria

A multipoint calibration is run to assess instrument linearity and establish response factors (RFs) used for quantitation. Multipoint calibrations are run following site start up, preventative maintenance, major repair or to correct the calibration of the instrument over time. At minimum, a multipoint calibration should be run once every six months. A four-point calibration curve is determined using mixtures of propane and benzene or toluene at concentrations of approximately 1, 2, 10, and 50 ppbV. The response factor is then calculated for each level for both propane or butane and benzene or toluene (depending on which one is the calibrant), then averaged together to provide the overall response factors for each species. The calculated response factor for propane or butane is applied to all compounds eluting on the PLOT column, and the calculated response factor for benzene or toluene is applied to all compounds eluting on the BP column. The multipoint calibration is considered acceptable by the operator when all four concentration levels are detected and the percent relative standard deviation (%RSD) of the four concentrations is less than **10**%.

- When a final RF is calculated following a previous RF from a multipoint calibration that is calculated after an adjustment or system change and no additional adjustments or system changes have been made to the analyzer, the affected ambient data is acceptable if the final RF (RF_{Final}) is within ±10% of the previous RF ($RF_{Previous}$).

$$Percent\ Difference\ (\%) = \frac{RF_{Previous} - RF_{Final}}{RF_{Final}} * 100$$

- If a multipoint calibration is run to correct system drift, and there has been no system change, or adjustment, it is not possible to back-apply a response factor. If the multipoint calibration has a greater than 10% RSD or percent difference, the site operator should immediately investigate the issue. **As long as QCs are passing, data remains valid**. If QCs are failing, refer to the appropriate chart for response actions. Refer to **Table 8** for Response Actions when a multipoint calibration does not meet the listed

Acceptance Criteria. Notify the operators and other personnel for all QC failures as necessary to determine if further investigation is warranted.

Table 8. PAMS Data Quality Objectives for a multipoint calibration used in the DAQ AutoGC program, including response actions for tests that are out of limits.

QC Check	Purpose	Frequency	Acceptance Criteria	Response Action
Multipoint Calibration	To assess instrument linearity and establish response factors used for quantitation	Prior to the start of sampling, following maintenance, major repairs, or calibration drift.	• The AutoGC operator is responsible for ensuring that the DQOs for determining response factors (RFs) are met according to the AutoGC Operator SOP (SOP UDAQ 41). • If a previous RF was set by the operator during instrument stabilization and no adjustments were made to the instrument between the system change and the acceptable multipoint calibration, then the final RF must be within 10% of the previous RF.	• Prepare new calibration. It may be necessary to investigate for system contamination or interferences resulting in suppression or enhancement of analytes. System leaks and trap degradation may impede a proper calibration as well as carryover from samples or standards. Improperly conditioned traps may contribute to chromatographic artifacts. PAMS data must not be reported unless calibration meets criteria. • If the final RF is >10% different than the previous RF, batch reprocess the affected data using the final RF back to the point of the system change. • If the recalculated CVS recoveries pass the CVS Acceptance Criteria (Table 2), then no further action is required. • If at any point the recalculated CVS recoveries fail, then invalidate the affected data from the first passing CVS back to the point of the system change.

4.3.8 Logbook and Checklist

Data validators review the operator logbook on a weekly basis at minimum or more frequently when quality control data or the site operator indicates there is an issue. The operator logbook and checklist are electronic logs kept by the site operator. The data validator is granted access to view the operator logbook and checklist via TeamViewer. Operator logbook entries typically summarize checks done on site or remotely, regular and preventive maintenance, problems, actions, repairs, and other items concerning instrument performance and data gathering. The operator

checklist is a list of required parameters that the operator must check weekly. Data in the checklist can reveal a variety of instrument malfunctions.

Upon monthly data validation, the data validator will copy relevant portions of the operator logbook and checklist to the monthly data validation checklist. Any entries that might impact the ambient data should be commented on and the corresponding chromatography reviewed.

Note: Before beginning the validation process for the ambient data, changes in RFs (as result of new multipoint calibration run), changes in the CVS, LCS or RTS calibration cylinders should be recorded in ORSATMAX.

4.3.9 Quicklook Reports

The primary tool that the data validator will use to review data are the daily Quicklook Reports located on ORSAT MAX. To access a daily Quicklook Report:

- From ORSAT MAX home, select the relevant AutoGC monitoring site from the map.
- On the "Monthly Data" tab, for each date, there are four buttons for each type of daily Quicklook report under the "View" column; alternatively:
- At the top right of the monitoring site's homepage, select the "Generate Quicklook" link.
- Select the relevant site, the relevant display, and the relevant date. It is recommended to leave "Minimum Display Value" and "Decimal" set to 0. Click the "Generate" button.

On a daily basis, there are some general items that the data validator should look for in any Quicklook:

- Missing hours which will display a pink status and chart button with "LST" in every compound cell. (NOTE: The hour after any missing hour is considered invalid due to the possibility of a collection error, such as the sample collecting in the thermal desorber for more than one hour. Apply the "AQ" null data code to the missing hour and the following hour and document it on the Monthly Data Validation Report (Form TAS-FORM-003) unless a more specific null data code

can apply. If six or more hours have been missing, apply the "AQ" or applicable null data code until the next passing CVS).

- Invalid hours automatically flagged which will display a red status and chart button. Sometimes invalid hours are accompanied by a secondary status color as an auto-flag for various issues (see the "Color Settings" for a legend).
- Investigate any over-range hours.

4.3.10 Retention Time Quicklook

The Retention Time Quicklook displays a chart of retention times for each compound for every hour of an entire day. Blank cells on the chart indicate the compound was not found or identified for that hour, or the retention time for the compound for that hour fell below the minimum display value that was entered in the Generate Quicklook query. The Quicklook also has a graphing function that can produce bar, line, and scatter graphs, editable color settings for hourly highlights, and the ability to generate a CSV of that hour.

The data validator will review the retention time Quicklook and take note of any inconsistencies. A retention time for a compound for one hour that differs significantly from the remaining hours that day may indicate a misidentification, aberrant chromatography, or a machine malfunction. Retention times that increase or decrease over time significantly may indicate flow issues or may simply be due to temperature and humidity variations in the ambient air.

If a data validator identifies an inconsistency that does not repeat significantly, the chromatography must be reviewed to determine if a correction must be made. If the inconsistency is repeatable, the method may need to be changed to reflect this. See **Figure 1** and **Figure 2** for an example retention time Quicklook.

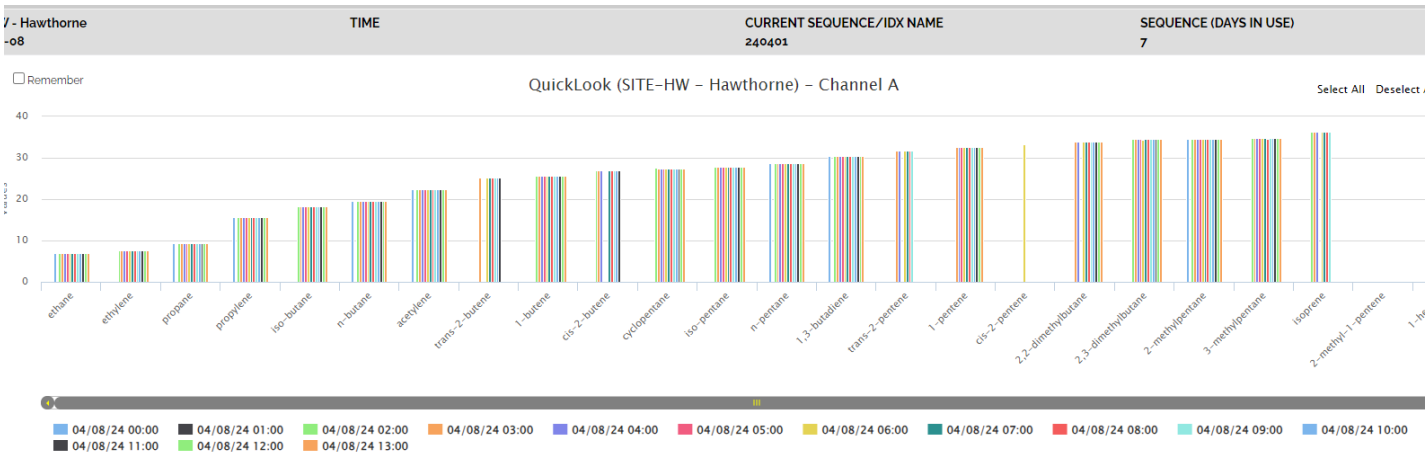


Figure 1. Example Retention Time Quicklook Report Graph

HR	STATUS	CHART	FILE	TIME	ETHANE	ETHYLENE	PROPANE	PROPYLENE	ISO-BUTANE	N-BUTANE	ACETYLENE	TRANS-2-BUTENE	1-BUTENE	CIS-2-BUTENE	CYCLOPENTANE	ISO-PENTANE	N-PENTANE
00:00			HWCD08A	00:47	7.0133		9.3700	15.7300		19.5133	22.3800						28.6233
01:00			HWBD08B	01:47													
02:00			HWSD08C	02:47	7.0167	7.6767	9.3700	15.7100	**18.2533**	**19.5133**	22.3167		25.5733	**27.0600**	**27.6233**	**27.7933**	28.6233
03:00			HWSD08D	03:47	7.0100	7.6800	9.3667	15.7033	18.2467	19.5100	22.3133	25.1200	25.5800	27.0400	27.4833	27.7900	28.6200
04:00			HWSD08E	04:47	**7.0200**	**7.6867**	**9.3733**	15.7100	18.2533	19.5133	22.3233		**25.5833**	27.0500	27.4767	27.7933	28.6233
05:00			HWSD08F	05:47	7.0133	7.6700	9.3667	15.7100	18.2500	19.5100	22.3133		25.5633		27.4833	27.7900	28.6200
06:00			HWSD08G	06:47	7.0167	7.6733	9.3733	15.7133	18.2500	19.5133	22.3033	25.1233	25.5767		27.4767	27.7933	28.6233
07:00			HWSD08H	07:47	7.0100	7.6733	9.3667	15.7067	18.2433	19.5067	22.3067	25.1167	25.5733	27.0500	27.4733	27.7900	28.6200
08:00			HWSD08I	08:47	7.0133	7.6767	9.3700	15.7100	18.2467	19.5067	22.3000	25.1233	25.5733	27.0400	27.4800	27.7867	28.6200
09:00			HWSD08J	09:47	7.0167	7.6833	9.3700	15.7100	18.2467	19.5100	22.3167	25.1267	25.5733	27.0533	27.4800	27.7900	28.6200
10:00			HWSD08K	10:47	7.0133	7.6800	9.3700	15.7167	18.2500	19.5133	22.3300	25.1300	25.5767	27.0567	27.4767	27.7900	28.6200
11:00			HWSD08L	11:47	7.0133	7.6767	9.3700	**15.7200**	18.2533	19.5133	**22.3367**	**25.1333**	25.5833	27.0567	27.4867	27.7933	28.6233
12:00			HWSD08M	12:47	7.0100	7.6733	9.3667	15.7133	18.2467	19.5133	22.3233		25.5767		27.4800	27.7933	28.6233
13:00			HWSD08N	13:47	7.0100	7.6700	9.3667	15.7167	18.2500	19.5100	22.3300		25.5800		27.4867	27.7933	**28.6267**

Figure 2. Example Retention Time Quicklook Report Data Chart

4.3.11 Amounts Quicklook

The Amounts Quicklook looks very similar to the Retention Time Quicklook except that it displays the amounts (in ppbC) for every compound per hour. In addition to features previously mentioned, the Amounts Quicklook will highlight the highest values for each compound for a day in red.

The data validator will review the Amounts Quicklook for inconsistent and very high values and document these hours for review at monthly data validation. If the data validator notices a change in the typical amounts for a compound, the data validator will review the method for potential integration problems. If problems are found, the data validator will contact the site operator and request a method change. See **Figure 3** and **Figure 4** for an example amounts Quicklook display

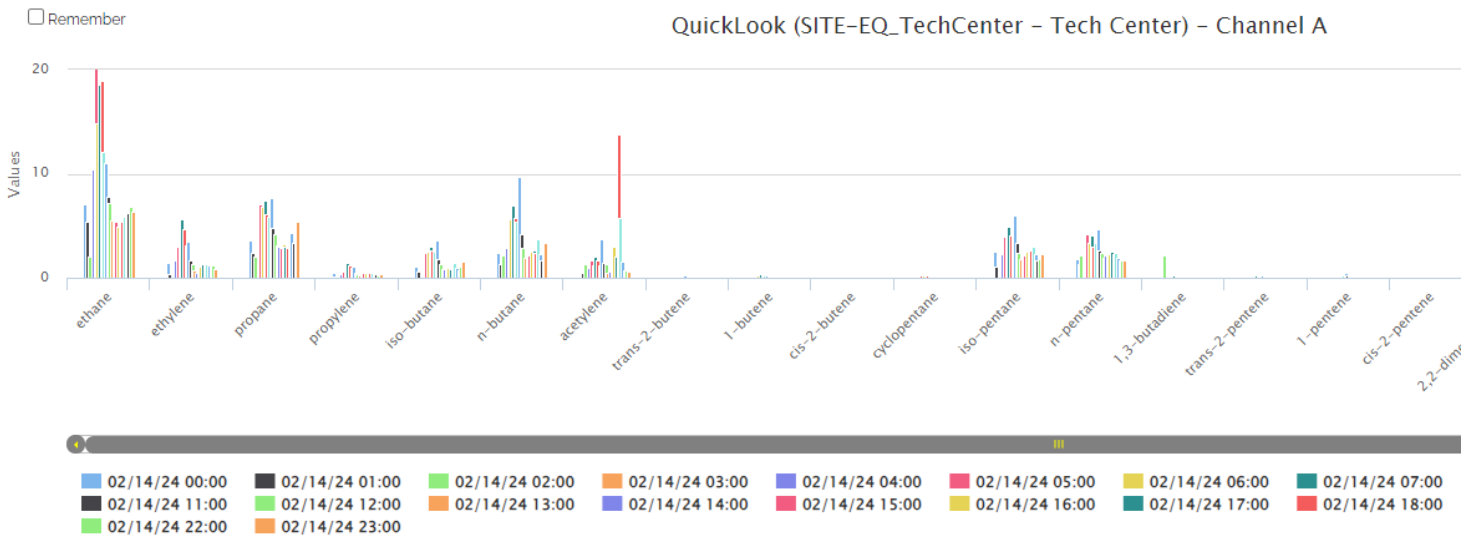


Figure 3. Example Amounts Quicklook Report Chart

HR	STATUS	CHART	FILE	TIME	ETHANE	ETHYLENE	PROPANE	PROPYLENE	ISO-BUTANE	N-BUTANE	ACETYLENE	TRANS-2-BUTENE	1-BUTENE	CIS-2-BUTENE	CYCLOPENTANE	ISO-PENTANE	N-PENTANE	1,3-BUTADIENE	TRANS-2
00:00			EQSB14A	00:47	7.0720	1.4672	3.5915	0.4994	1.0685	2.3763	0.6413	0.0424	0.1191	0.0159	0.0995	2.4710	1.8449	0.1262	0.0317
01:00			EQSB14B	01:47	5.3659	0.3922	2.4291	0.1025	0.6443	1.3185	0.4395		0.0393		0.0376	1.1210	1.2723		
02:00			EQCB14C	02:47	2.0623		1.9875			2.1695	1.3536						2.1335	2.1483	
03:00			EQBB14D	03:47															
04:00			EQSB14E	04:47	10.3663	1.6412	5.5904	0.3926	1.5267	2.8198	1.0088		0.0632	0.0275	0.1700	2.2434	2.3092	0.0583	0.0609
05:00			EQSB14F	05:47	**20.1676**	3.0012	7.1047	0.6458	2.4078	5.6799	1.6500	0.0947	0.1570	0.0788	0.2174	3.9309	4.1501	0.1068	0.1369
06:00			EQSB14G	06:47	14.8541	3.1556	6.7680	1.0701	2.5351	5.6019	1.2491	0.1134	0.2221	0.1007	0.1956	4.4476	3.3512	0.1352	0.1871
07:00			EQSB14H	07:47	18.5984	**5.6841**	7.4390	**1.4094**	3.0042	6.9967	2.0003	**0.2126**	**0.3374**	**0.1466**	**0.2451**	4.9443	4.0642	**0.2307**	0.1837
08:00			EQSB14I	08:47	18.8736	4.7033	6.1483	1.2103	2.6062	5.6957	1.6259	0.1350	0.2648	0.1070	0.1819	4.0357	3.0483	0.1644	0.1326
09:00			EQSB14J	09:47	12.1329	3.1103	5.8575	0.9932	2.4625	5.3317	1.2458	0.1004	0.2313	0.0782	0.1577	4.0687	3.3938	0.1359	0.1078
10:00			EQSB14K	10:47	11.0493	3.4186	**7.6850**	1.0511	**3.6403**	**9.7592**	3.6699	0.1883	0.2500	0.1125	0.2105	**5.9424**	**4.6420**	0.1298	**0.2583**
11:00			EQSB14L	11:47	7.7687	1.7087	4.7800	0.5274	1.7367	4.1677	1.4781	0.0554	0.1248	0.0479	0.1191	3.3176	2.6020	0.0632	0.1026

Figure 4. Example Amounts Quicklook Report Data Chart

4.3.12 Email Notifications Sent by AutoGC Data Validators

Constant oversight and immediate communication of instrument or communications issues potentially affecting data is critical for resolving issues quickly and minimizing data loss. Documentation of large periods of data loss and their causes are essential as they affect data recovery.

- If an on-going or otherwise unresolved instrument or data problem is identified, the data validator will send an email to the site operator(s) and AMS section manager describing the problem. The data validator will continue to monitor the problem and send necessary emails until the problem is resolved.
- When one or both columns have had 336 continuous hours or more of invalidated or lost data, the data validator will notify via email the site operator(s), and the AMS section manager. The data loss period, reasons for the data loss, and steps taken to resolve the problem will be documented in an MDVR and Memo.
- When a consistent over-range value is determined to be ambient and valid, the data validator will send an over-range data notification email to the site operator, and AMS section manager including the date/time, compound, concentration, and a wind rose for the site at the affected hour.
- The data operator is responsible for sending email notifications for changes in the CVS, RTS, or LCS stock cylinder, CVS blend ratio, and LCS dilution factor to the monitoring PAMS coordinator who will be updating the information in ORSAT MAX. The data validator will confirm the data is uploaded into ORSAT MAX and document the installation on the MDVR. Documentation will be saved

U:\PLAN\AMC\GC-Agilent+Markes\Data\MDVR\Cylinder documentation

4.3.13 Preparing the Monthly Data Validation Report and Assembling Data (MDVR)

- Ensure that hours with no data in ORSAT MAX do not have associated data files (if there is data, load it manually into ORSAT MAX).
- Ensure that all operator and validator-reprocessed data has been uploaded into ORSAT MAX.

- Create a data validator notes file to document information and changes made to the data with explanations as to why the changes occurred.
- Using the Quicklook reports and other available data, select what hours will have their chromatography reviewed by highlighting the date/hour boxes for each column on the hourly grid charts of the MDVR. Hours required to be reviewed are:
    - All quality control hours;
    - The first and last ambient hours of the month;
    - The ambient hours with the highest TNMHC per column per day;
    - Any hour with the highest daily value occurring for five or more compounds on one column;
    - Hours before and after invalidations, corrections, and batch reprocesses (per column; for example, if a compound on the PLOT column is invalidated for an hour, only the PLOT column for the hours before and after the hour needs to be reviewed);
    - Any hour highlighted in the grid boxes even if inadvertently;
    - A minimum of three ambient hours each day.
- The data validator will select the remaining hours based on the following guidelines:
    - Select a representative sample of hours based on weather conditions, trends in ppbC amounts, and trends in retention times;
    - Select hours where an individual compound is much higher than the other hours that day (the data validator shall use best, consistent judgment);
    - Select hours with missing compounds that may normally be present;
    - Select hours with unusually high unidentified ppbC totals;
    - Select hours with unusual trends across multiple hours or compounds;

- Generally, no more than six hours per day should be selected by the data validator for review; however, this may be exceeded in unusual circumstances.

4.3.14 Reviewing Communications with the Operator

- Retrieve relevant operator logbook and checklist entries from the relevant Google Docs/Sheets files.
- Read the operator logbook and checklist entries and make notes in the data validator notes file to reflect any changes or issues that may affect the validity of the data.
- Read any email correspondence collected for the site/month and make notes in the data validator notes file to reflect any changes or issues that may affect the validity of the data.

4.3.15 Examining Quality Control Charts

- Retrieve the quality control charts for the site/month from ORSAT MAX.
- Examine the quality control reports and compare the results against the acceptance criteria outlined in Table **2** through **Table 5** and **Table 7** and **Table 8**. If all quality control results are within the acceptance criteria, determine whether or not the failure was due to a causative event.
- If a failure is due to a Type II causative event, invalidate the affected quality control hour with the appropriate flag. Record the flag, hour, and a brief description of the cause for the failure on the MDVR.
- If the quality control failure is not due to a Type II causative event, all affected ambient and quality control data must be invalidated by applying the appropriate flag codes.

4.3.16 Applying Flag Codes for Out of Limit Quality Controls

Table 2 through **Table 5** and **Table 7** and **Table 8** summarize the quality control checks run on the AutoGCs, including their frequency, acceptance criteria, and response actions required. A complete list of qualifying flags can be found in Appendix B.

If the recovery of any calibrant falls outside its limits, then it will be highlighted red in the quality control summary chart in the Quicklook reports.

- The entire column of the calibrant species when it fails the CVS.
    - If the percent recovery for calibrant species is not within **70-130** %, then the respective column is considered invalid backward and forward to a valid CVS in which the calibrant recovery for each column was within limits. Unless technical justification is provided to explain nonconformance, minimally qualify as "QX" or apply the "AS" null data qualifier to both the PLOT or BP columns.
    - If the CVS failure is reflecting an overarching problem that is not specifically related to propane and benzene recoveries, then a more appropriate null data qualifier may be applied to both columns.
    - Document the column failure(s) in the data validator notes and record all invalidation information in the MDVR.
- Acetylene Failure in the CVS
    - If the percent recovery for acetylene falls outside of the 70-130% limit in the CVS, qualify or invalidate only the respective compound that failed backward and forward to valid CVS hours in which it passes.
    - Apply the "EC" or "AS" null data qualifier or the "LJ" qualifier and record all invalidation/qualifier information on the MDVR.
- Part of a column is invalid when a calibrant (other than propane or butane, Toluene, acetylene) fails the CVS.
    - If the percent recovery for a calibrant other than propane, benzene, acetylene falls outside of the 70-130% limit, qualify or invalidate that particular calibrant and every target compound, by elution order, after the previous passing calibrant up to but not including the next passing calibrant backward and forward to calibrant-passing CVSs.
    - If eight or more compounds are invalidated for a single hour due to multiple calibrant failures on the CVS, invalidate TNMTC and TNMHC for the same time range.

  - Assign the "EC" or "AS" null data qualifier to all affected target compounds and record all invalidation information in the MDVR.

- Dean's Switch ("Cut-Time") Errors
  - For any QC or ambient hours in which a compound is partially or completely excluded from the PLOT or BP columns due to a Dean's Switch ("cut-time") error, apply the "AQ" null data qualifier to only the affected compound(s) from when the problem began until it is resolved. The most commonly affected peaks are isoprene, 1-hexene, and n-hexane.

- CVS Failure Due to Calibrant Misidentification
  - When a calibrant fails due to misidentification or misintegration, review the CVS chromatogram along with those of the surrounding ambient hours.
  - If the misidentification is limited to the CVS hour, correct the misidentification following the steps in **Table 2**.
  - If the target is also misidentified in the adjacent ambient samples, invalidate, qualify, or reprocess the affected data as outlined in **Table 2**.
  - To invalidate individual compounds due to misidentification or misintegration, apply the "BH" null data qualifier.
  - Document all flagging and batch reprocess information in the MDVR.

- System Contaminant in the Blank
  - When TNMHC is greater than 10 ppbC in a blank that has neither a Type I nor Type II causative event, the blank fails and TNMHC and TNMTC are invalidated. Apply the "SC" null data qualifier from the previous TNMHC-passing blank to the next TNMHC-passing blank.
  - When TNMHC is greater than 10 ppbC in a blank that has a Type I causative event, the blank fails and TNMHC and TNMTC are invalidated. Apply the "SC" null data qualifier from the causative event to the next TNMHC-passing blank.
  - If a blank failure occurs due to the known network-wide PLOT system contaminant isobutylene, the PLOT method should be adjusted by the operator or validator to de-

integrate this peak and exclude it from TNMHC, then reprocess the data and upload it to ORSAT MAX. Alternatively, TNMHC and TNMTC can be invalidated by the data validator by applying the "SC" null data qualifier as described above.

- When the blank method incorrectly integrates an electronic spike causing the blank to fail for TNMHC, adjust the method (session 4.4) for the affected hour(s) to de-integrate the electronic spike, then reprocess the data and upload it to ORSAT MAX.
- Document all flagging and batch reprocess information on the MDVR.

- Individual Compounds in the Blank
  - When an individual target compound is above the MDL will be qualified with the "QX" flag, or nullified with the "AS" flag. The flag will apply to that compound backward and forward to passing blanks. The compound will be highlighted red in the blank summary chart on the Quicklook report on ORSAT MAX.
  - Document all flagging information on the MDVR.
- Missing Compound(s) in the Retention Time Standard
  - When a compound is not identified in the Retention Time Standard, it is usually a result of misidentification, misintegration, non-integration, or lack of presence in the standard.
  - If a missing compound occurs in the RTS, correct the RTS chromatogram (session 4.4) for the misidentified target compound(s) and request the operator to optimize the method for future AutoGC runs as necessary.
  - Missing compounds in the RTS should alert the data validator to the possibility of a misidentification in the ambient samples. If the problem is found to exist in the surrounding ambient data, invalidate, correct, or ask the operator to reprocess the affected data.
  - To invalidate individual compounds for misidentification, misintegration, or non-integration, apply the "BH '' null data qualifier to the affected period.

  - Document all flagging and batch reprocess information in the MDVR.

- Audit Samples
  - When an audit sample is run, apply the “AZ” null data qualifier to the audit sample and the following blank.
  - Document all flagging information on the MDVR.
- Multipoint Calibration
  - When a multipoint calibration is run, flag the calibration hours and the following blank(s) as “AT”.
  - If the last blank following the multipoint calibration fails, flag in accordance with the procedure listed in the “Individual Compounds in the Blank” section above.
  - Document all flagging information on the MDVR.

4.3.17 Applying Flag Codes for Non-QC Data

- Minimum Detection Limit (MDL) Studies
  - When a MDL study is run, flag the MDL study and surrounding blanks as “DL”.
  - Document all flagging information on the MDVR.
- Machine Malfunction
  - When a machine malfunction (such as a trap failure) occurs that cannot be ascribed to a more specific qualifier, apply the “AN” null data qualifier.
  - Document all flagging information on the MDVR.
- Air Conditioner/Heating Failures
  - When the shelter temperatures exceed or fall below the operating range of the AutoGC due to failure of the trailer air conditioner or heater, apply the “AE” null data qualifier from the beginning of the problem until the next passing CVS.
  - Document all flagging information on the MDVR.
- Inclement Weather or Catastrophic Site Events

  - When data is missing or invalid due to instrument issues associated with inclement weather, such as severe thunderstorms, or site wide catastrophic events, apply the "AO", "AD", "AP", "AW", or "AM" null data qualifiers, dependent on the exact cause (see Appendix for a description of each qualifier) to all invalid or missing data from the beginning of the problem until the next passing CVS.
  - Document all flagging information on the MDVR.
- System Timing Errors
  - When the AutoGC is collecting less than 75% of the data within the hour assigned, apply the "AG" null data qualifier from the beginning of the problem until the next passing CVS.
  - If the time zone on the site computer is wrong, review the Acquisition Time in the tx1 files. If the acquisition time is also wrong and needs to be adjusted, the data is invalid, and the "AQ" or "AG" null data qualifier must be applied from the beginning of the problem until the next passing CVS. If the acquisition time is correct and does not need to be adjusted, the data is valid.
  - Document all flagging information on the MDVR.
- Missing Hours
  - A variety of issues not previously noted can cause missing data. If the AutoGC is down for one to five hours due to an event that does not affect the AutoGC calibration, apply to only the affected hours and the hour collected following the missing hour(s) the "AF", "AI", "AQ", "AV", or "BK" null data qualifiers dependent upon the exact cause. If the cause is unknown, apply "AM". Any ambient hours collected between the flagged hours and the next scheduled CVS should be examined carefully for any indication of aberrant chromatography. If the chromatography is aberrant in any way, include the affected hour(s) in the flagging block.
  - If the AutoGC is down six or more hours, apply the appropriate null data qualifier previously described from the start of the missing hours until the next passing CVS. If the next scheduled CVS fails without a Type I or Type II

Causative Event, flag the data backward and forward as outlined in **Table 1**

- Document all flagging information on the MDVR.

- Preventive Maintenance, Major Repairs, and Recalibrations
  - After annual preventive maintenance, major modifications, repairs, or when the system has drifted out of limits as indicated by a failing or near failing CVS and/or LCS, the system must be recalibrated.
  - A multipoint calibration will be performed by the operator and used to accurately determine the system's response factors.
  - Apply the "BA" null data qualifier to any maintenance or repair period with respect to the instrument. Apply the "BE" null data qualifier if the maintenance or repairs affect the entire site. Include the period of ambient data following maintenance or repair up to the next passing multipoint calibration or CVS. Apply the "AT" or "BC" null data qualifier to the multipoint calibration following the maintenance or repair period.
  - Document all flagging information on the MDVR.
- Over-Range Values
  - An over-range value is a value that is applied to peaks that are too high for EZChrom to compute.
  - Concentration values in the *.tx1 file will appear as dashes or asterisks, and the value may not be visible in ORSAT MAX.
  - To verify the validity of the data:
    - View the chromatography of the hour and surrounding hours
    - Examine the operator logbook and check sheet for the site
    - Verify the QC checks before and after the over-range hour
    - Compare data to other regional monitors, historical data, and meteorological data if available

        - Communicate with the operator for any events at the site that may have affected the data.
    - If the over-range designation is due to electronic spiking or other aberrant reasons, either apply the "DA" null data qualifier for aberrant data or manually de-integrate the peak after ensuring the quality of other data during that hour. Save the data and reload the *.tx1 file into ORSAT MAX.
    - If the data is confirmed as ambient and valid:
        - Flag data with the "EH" Estimated; Exceeds upper range flag in ORSAT MAX.
        - Upload the reprocess *.tx1 file into ORSAT MAX and verify the calculated value.
        - If the over-range value is consistent send an over-range email notification as outlined in Section 4.3.12.
- Other Events
    - As outlined in Appendix B, there are numerous types of EPA qualifying flags available.
    - The data validator shall use best judgment to determine what flag may be most appropriate for an event.
    - Not all flags will be used. If the data validator believes that either a flag that has not been used before or a very generic flag is most appropriate for an event, the data validator shall consult with others to ensure that it is the one that should be used.
    - Document all flagging information and reasoning on the MDVR.

4.3.18 Verifying Chromatography Data Files

Ensure that all chromatography data files have been extracted from their respective Daily-ZIP folder into the data validator's relevant AutoGC Data folder. If there have been any operator or data validator-reprocessed data files, ensure they are extracted last so they overwrite the original files in the data validator's AutoGC data folder.

4.3.19 Accessing Chromatography in EZChrom

- Open the Agilent Technologies Control Panel either via the Start Menu or a Desktop Icon.
- Enter your username and password. This should be set up upon installation.
- Navigate to the "Instruments" tab. Under "Start Instrument", click the "Project" selector and select the relevant site that requires validation as shown in **Figure 5**.
- Click "Launch Offline". This will open the EZChrom window.

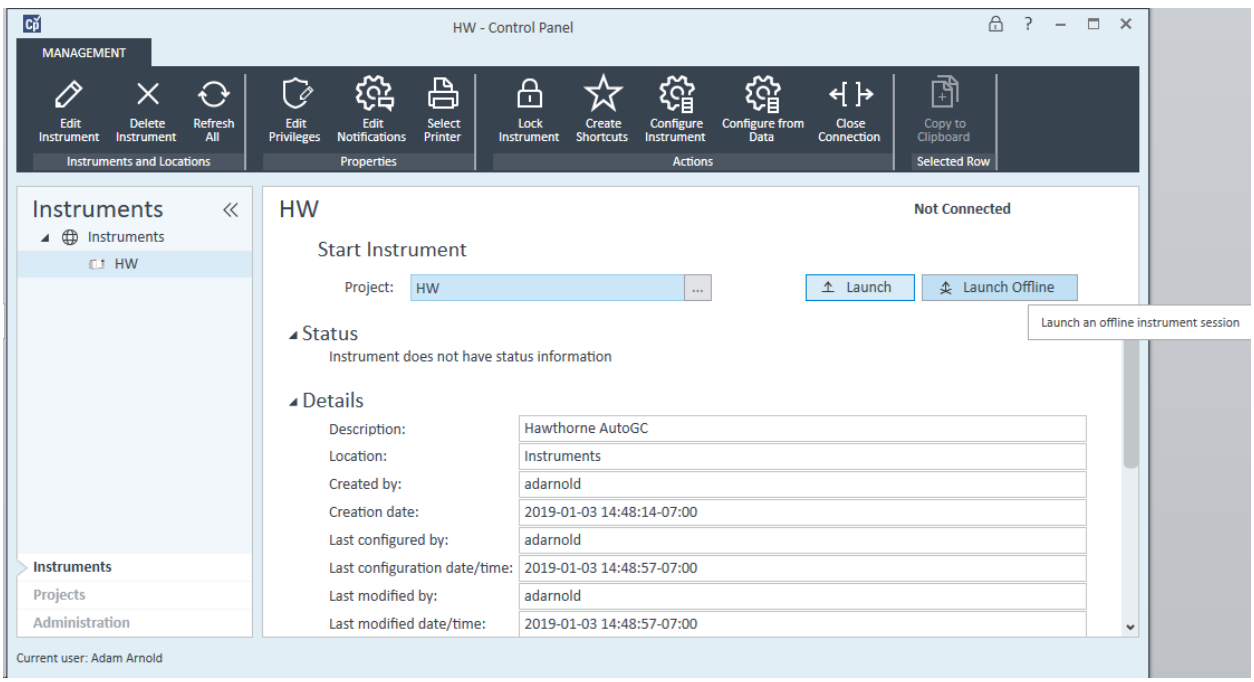


Figure 5. Agilent Control Panel Instrument Display

- EZChrom will usually open where the user last left off if nothing was closed prior to it being shut down previously.
- The data validator may necessitate the creation of a validating sequence prior to reprocessing the data files. To do this:
  - Click the "Instrument Wizard" button .
  - Click "Create a sequence".

- Select the appropriate method in the "Method" box. This should be present in the AutoGC data file after extracting from the Daily-ZIP files.
- Under "Data File Type" select "From existing data files". Click "Next".
- Click the "Open Files" button . Navigate to the AutoGC Data folder with the relevant month's data, and select all of the .dat files for that month. Click "Open".
  - Only five or six days of files can be added to the sequence at a time. The last file added will appear as the first file of the files added. To add more files than the initial group of files, repeat the process of selecting and opening files until all of the desired files appear in the sequence.
- Click "Finish".
- Go to File -> Save As -> Sequence. Navigate to the relevant month's AutoGC data folder. Save the new sequence as *RPO startdate to enddate"SiteID"*.seq. Be aware that some symbols are not allowed in the sequence name.

- After creating and saving the sequence, it will need to be processed as shown in **Figure 6**:
  - Click the "Process Sequence" button .
  - Ensure the "Sequence Name" file is the current sequence that has already been saved.
  - Ensure that the "Result path" matches the AutoGC data folder that contains the data for that month.
  - For "Result Name" use the same format as the sequence name *RPO startdate to enddate "SiteID"*.rst.
  - Ensure that under "Run Range" the "All" radio button is selected and everything under "Report" is blank or grayed out.
  - Under "Mode", "Tower" should be grayed out, "Use last results" should be selected under "Processing", and "None" should be selected under "Bracketing".
  - The "Review" section should be blank or grayed out.

- Click “Start”.
- The program will spend up to a few minutes opening each .dat file to verify its contents.
  - To ensure quicker processing of the sequence, be sure that no overlays are active in either column view. Each overlay will be rendered again for each hour of the sequence processed and that can take a lot of extra time.

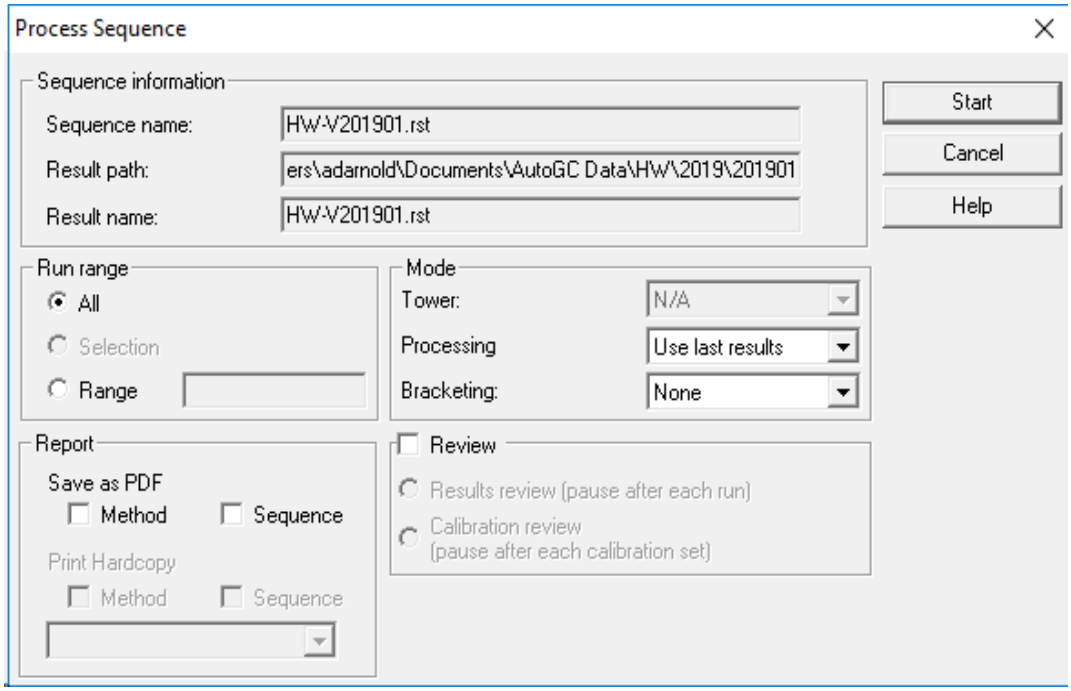


Figure 6. Process Sequence Dialog Box

- Once the sequence has been processed, a yellow flag saying “Result Review Mode” (RRM) should appear in the upper toolbar.
- In RRM, hourly data files can be easily accessed by navigating to the “Result Sequence” window and double clicking on a .dat file under the “Filename” column.

4.3.20 Reviewing Chromatography in EZChrom

- Beginning with an open hourly data file as previously described, begin by ensuring that the relevant column that is being reviewed is selected. There is a drop-down menu for the signals in the upper toolbar. “1:Front Signal” is the PLOT column, and “2:Back Signal”

is the BP column. The proper signal must be selected before reviewing.

- Right click on a chromatography window and click “Annotations”. Ensure under the “Peaks” drop down menu that the following parameters are present under “Show the following annotations”: Name and Retention Time. More may be selected, but note that the more that is selected, the more cluttered the chromatography will appear. Under “Other”, make sure the following are checked: Baseline, RT Window, Show undetected named peaks. Click “Apply to All” and “OK”. This may have to be done for each signal.
- Chromatography colors and fonts may be changed. Refer to the EZChrom help file or users’ manual for further instructions.
- For each required hour to be reviewed, visually inspect chromatography. Keep the following tips in mind to troubleshoot, repair, and document abnormalities seen in chromatography:
    - Because of differences in humidity, temperature, and instrumentation, compare the identified location of the target peaks against the location of the corresponding peaks in the RTS, CVS, and/or LCS collected from the same site. Look for problematic compounds such as 2-methylpentane, 2,3-dimethylbutane and 3-methyl pentane in the PLOT column.
    - Use peak patterns and relative retention times to determine if the target compounds are identified correctly in the ambient air samples.
    - Look for discrepancies including but not limited to misnamed samples, electronic spikes disrupting the chromatography or affecting target peaks, samples with obvious baseline abnormalities, poor or inconsistent peak integration, peak misidentification, peaks co-eluting, and samples with cut-time errors.
    - Use null data qualifiers to invalidate individual compounds, whole columns, or whole aberrant hours. “BH”, “AN” and “DA” will be most commonly used.
        - “AM” should be used sparingly and only with justification listed in the MDVR.
    - Evaluate quality control reports, operator logbook and checklist entries, and correspondence with the operator in

concert with the examination of corresponding chromatography to determine the actual start and stop time for flagging.

4.4 Correcting Chromatography

Chromatography needs corrections applied on occasion. To determine if a chromatogram requires corrections, follow **Figure 7**.

Corrections to chromatograms can be applied two ways. Individual hours can be corrected (Section 4.4.1), or multiple hours can be corrected at once by creating a method and applying a batch reprocess (Section 4.4.2).

Individual hours requiring corrections to chromatography must be noted in the MDVR hourly grid chart and in the chart of corrected chromatography. Batch reprocesses should be noted in the MDVR batch reprocess section. Both the original and final reprocessed files should be in the event that the corrections needed to be reverted back to the original.

4.4.1 Minimum Correction Threshold (MCT)

To avoid excessive amounts of time performing corrections to the chromatography, a MCT is used to determine the minimum amount required for a correction to be saved or reprocessed manually. The MCT is defined as the larger of 1.0 ppbC or 10% of the original peak size.

The original peak size is determined by the ESTD concentration displayed on the chromatography. If the peak in question is below the MCT, no adjustment will be made. If the difference between the original peak size and the new peak size is below the MCT, as determined by making the proposed correction, clicking “Analyze”, and reviewing the new ESTD concentration, the change can be saved with documentation in the data validator notes if another correction requires the file to be saved.

If the difference between the original peak size and the new peak size is below the MCT, and no other correction in the file requires saving, the change is not saved.

If the difference between the original peak size and the new peak size is at or above the MCT, then the modified .dat file is saved and a new *.tx1 file is generated (see Section 4.5.1).

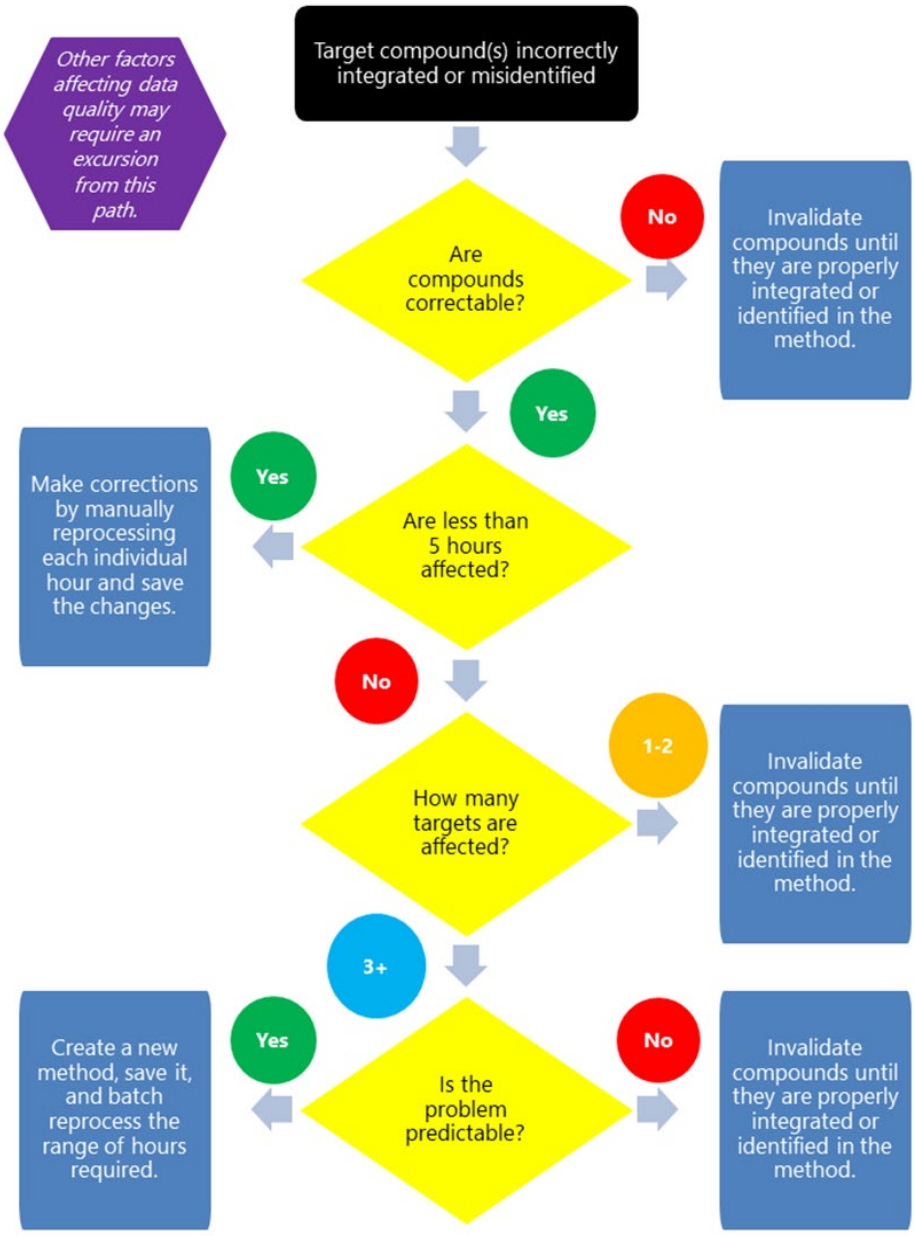


Figure 7. Chromatogram correction decision flowchart

4.4.2 Applying Chromatogram Corrections to Individual Hours

When reviewing chromatograms in EZChrom make chromatogram corrections following the MCT discussion in Section 4.4.1 and the

flowcharts in **Figure 7**. If a large number of hours require a consistent correction, correct the chromatograms as outlined in the batch reprocess discussion in Section 4.4.3.

The following functions are the most commonly used while correcting chromatography. The data validator may use others when necessary. See the EZChrom Help menu for more information on all integration functions that can be performed in EZChrom. Most of these functions can be found on the integration toolbar at the bottom of the EZChrom screen.

- Adjust Retention Time Window: Used for shifting the retention time windows used to identify peaks. This can be used to correct the identification of peaks.
- Reassign Peak: Used as an alternative function for correcting the identification of peaks.
- Manual Peak: Used to manually correct the integration of an existing peak.
- Manual Baseline: Used for manually shifting the baseline anywhere on the chromatography.
- Split Peak: Used to split a co-elutant from a mother peak with a straight split.
- Exponential Skim/Front Exponential Skim: Used to split a co-elutant from a mother peak exponentially.
- Disable Peak End Detection: Used to prevent an unwanted split in a peak.
- Integration Off: Used to prevent integration of any peaks for a range in the chromatogram.
- Width: Value used to calculate bunching/smoothing factors; higher numbers will reduce the amount of data points across a peak as too many data points can cause faulty integration, but values too high can result in peaks not being integrated. Can be applied to whole chromatogram or small parts.

- Threshold: Value used to distinguish the start and stop points of chromatogram peaks from baseline noise and drift; higher numbers can result in a reduced area of peak integration. Can be applied to whole chromatogram or small parts.
- Minimum Area: Value used to set a minimum limit for a peak to be integrated; higher numbers will result in more peaks being eliminated as noise.

There are more integration functions available in EZChrom than listed here. The data validator may use these other functions when necessary; review the EZChrom Help function for further descriptions on how to use them.

When an integration correction is made, a dialog box will appear confirming the retention time location of the correction and presenting several options (See **Figure 8**).

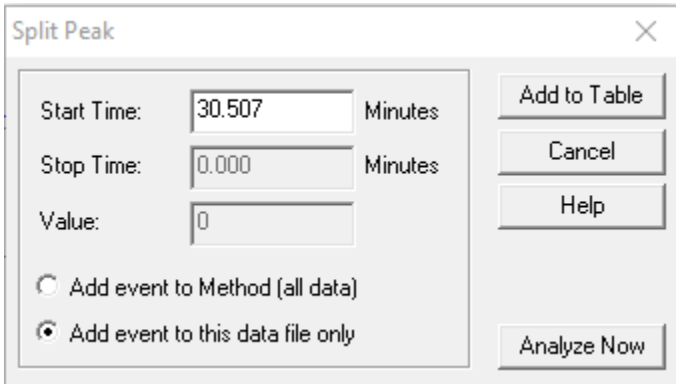


Figure 8. Example Integration Correction Dialog Box

For the purposes of correcting individual hours, select the radio button next to “Add event to this data file only”. To instantly perform the chromatogram correction, click the button “Analyze Now”. If “Add to Table” is selected, EZChrom will add the event to the manual integration fixes table, but will not perform the correction until Analysis -> Analyze is selected. Another dialog box may appear confirming the method change and if it should be saved; click “Yes” to ensure the correction is performed.

Once all corrections have been made, save the data file and produce a new *.tx1 file by reprocessing the result sequence:

- Go to Result Set -> Process.
- Verify the Sequence name, Result path, and Result name are accurate. If they aren't, the sequence will need to be reprocessed separately outside of Result Review Mode.
- Select the Range that refers to the sequence line of the relevant corrected data file.
- Leave all check marks under "Report" blank.
- Under "Mode" select "Reintegrate" for Processing and "None" for Bracketing.
- Press "Start".
- Proceed to reloading corrected *.tx1 file(s).

4.4.3 Applying Chromatogram Corrections to Multiple Hours

Sometimes there is an ongoing method issue affecting many hours which would require the same corrections across all affected hours. This is called a batch reprocess. A typical example is having the method shift on the PLOT column due to flow or temperature issues that cause the misidentification of several compounds.

Following the flowchart in **Figure 7**, if a batch reprocess is required:

- Select a representative data file during the period with an approximate average displacement of retention time windows (likely an hour with an average of temperature and humidity over the period).
- Save the method as a new file by going to File -> Save As -> Method. The filename format must include the Site ID, year and month at a minimum where the date for the filename represents the first date of the data requiring reprocessing.
- Perform the necessary corrections utilizing the tools outlined in Section 4.4. When the dialog box in **Figure 8** appears after a correction is made, select the radio button for "Add event to Method (all data)" before clicking "Analyze Now". Another dialog box may appear confirming the method change and if it should be saved; click "Yes" to ensure the correction is performed. If a method audit trail window pops up, add sufficient reason for adjusting the method.

- When all corrections are complete, save the method again by going to File -> Save -> Method.
- In the Result Sequence, for the range of hours where the problematic method has been identified, navigate to the "Method" column and change the method to the newly created validator method to be used for batch reprocessing.
- Reprocess the range of data files that need to be corrected by going to Result Set -> Process. Ensure that the radio button next to "Range" is selected with the proper sequence line range for the data files that need to be corrected entered. Ensure that "Reintegrate" is selected next to "Processing", and "None" is selected for "Bracketing". Ensure all checkboxes under "Report" are unchecked. Press "Start".
- Once the batch reprocess is complete, review the corrected files to ensure that the required corrections have been made for all data files.
    - Review all reprocessed QC files and at least 3 ambient hours a day.
    - If there continue to be many problems across many files, select a different data file and re-correct the method following the steps above. Reprocess and re-review.
    - If 1-2 compounds are still misidentified, invalidate those compounds.
    - If less than 5 hours continue to have multiple problems, correct those hours manually following the steps in Section 4.4.2 and reprocess. Do not apply these changes to the method.
    - If the change is correct for 5 or more hours initially, but the method continues to drift and is consistently wrong for many hours, a second, new method may need to be created, and the steps above should be repeated for those hours with errors.
- Once the corrections have been verified, document the batch reprocess performed and the new method created in the batch reprocess section of the MDVR.

4.4.4 Reloading Corrected Files

- After completing all corrections and batch reprocesses, ensure all corrected hours have a new modified date for *.dat and *.tx1 files.
- Navigate to the ORSAT MAX site page.
- On the left toolbar of the ORSAT MAX page, click on the Data Manager Icon -> Upload EZChrom.
- Click “Select Data Files”.
- Navigate to the relevant AutoGC Data folder and select the corrected *.tx1 files. Click “Open”.
- Select the checkbox next to “Update Data”. Failure to do this will result in the data from the corrected *.tx1 files not being properly uploaded.
- Click “Start”.
- Once the upload process is complete, navigate to a amount or retention time Quicklook page for a corrected hour (depending on what correction was performed) and verify that the data matches the corrected *.tx1 or *.dat files.

4.5 Data Validation Using MAX

Each trained data validator has the responsibility for their review, validation, and verification of the measurements from assigned sites. Technical staff, including the data validator, will review data summaries and investigate data outliers and unusual data.

Each data validator maintains records of all activities, facts, and follow-up actions relating to flagging of ambient data to include: electronic validator notes, emails, and other records as appropriate. Archive any electronic correspondence. The detail of record keeping should be sufficient to reconstruct the data validation event at a later time without having to rely on memory or handwritten notes that may not be readily available.

The data validator will review and question any part of the measurement process. If the data validator finds an issue they believe to not follow the Data Quality Objectives or process, they will notify the site operator, technical/quality assurance coordinator, and/or the section manager as appropriate. Data validators may also initiate data reviews or other corrective actions to bring the process into compliance if necessary.

4.5.1 Single Day Screening & Validation

The validation option in ORSAT MAX is only available for people responsible for data validation. The tool can be accessed by clicking the check mark option (**Figure 9**)

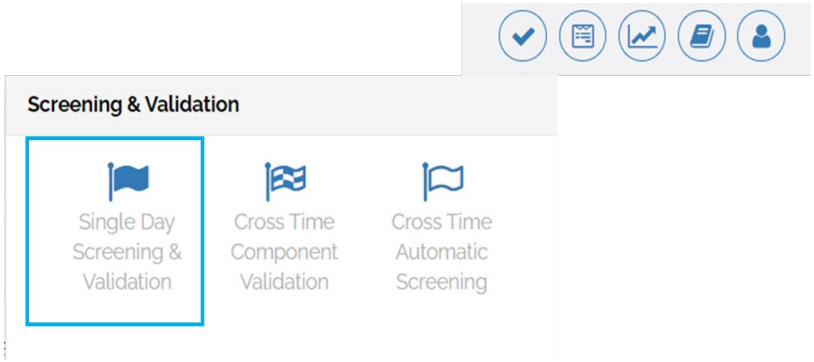


Figure 9. Screen & Validation option in MAX. The single Day Screening & Validation option highlighted in blue

- Select Site (HW), Date and "GO"

4.5.2 Flagging hourly data in MAX

- On the "Single Day and Validation" option, targets can be flagged for individual days by selecting the column for the chosen component (click in the box at the top of the column) and then using the edit icon to enter the desired code and select "Apply" (**Figure 10**)

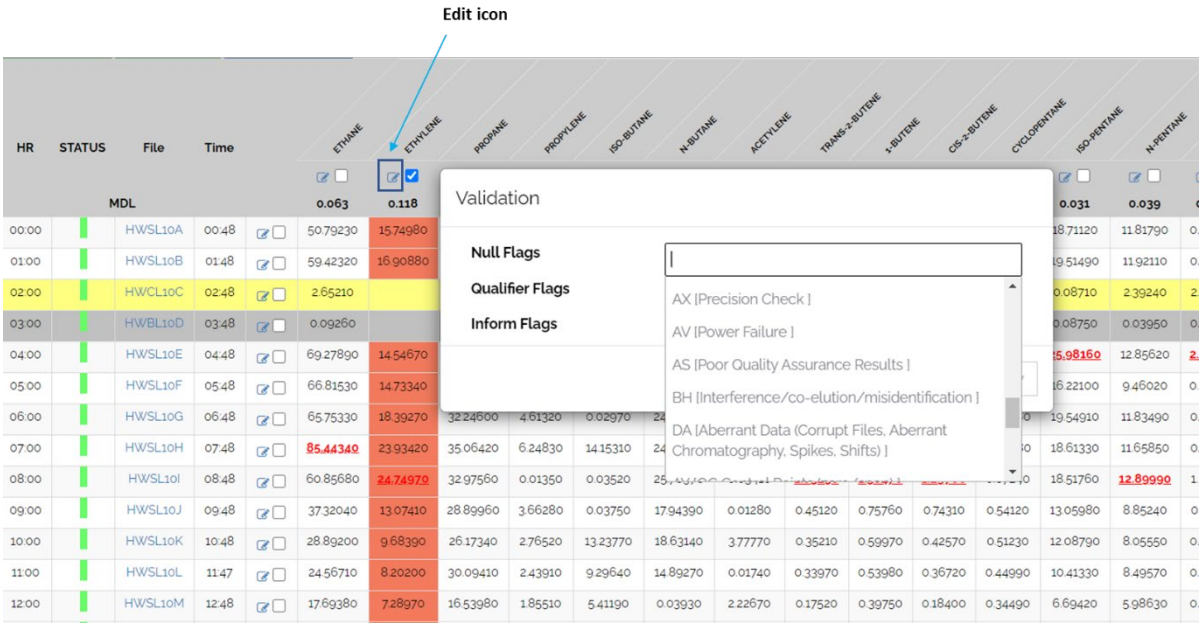


Figure 10. Flagging hourly data in MAX

- Individual target components can be Auto Screened for the day by selecting "Screen Qualifiers". A single day's data can be flagged for "Non Detect, ">= MDL and < SQL", and "Highest Point Cal" (**Figure 11**)

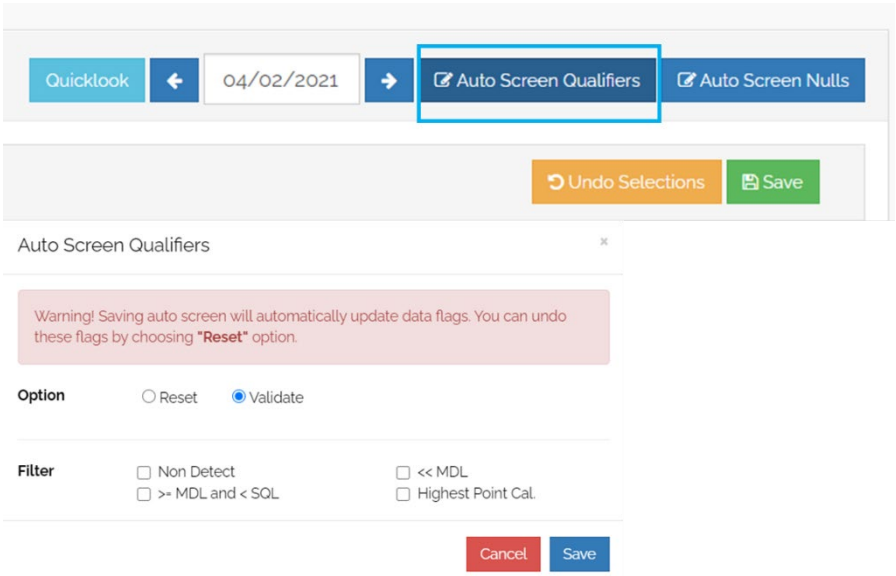


Figure 11. The Auto Screen form in MAX

- Individual target components can be flagged for a single hour by clicking in the selected cell (compound/hour). The selected hour will be highlighted in red and the validation pop-up menu will pull up when right clicking the mouse. Then choose the desired code from the dropdown menu and select "Apply" (**Figure 12**)

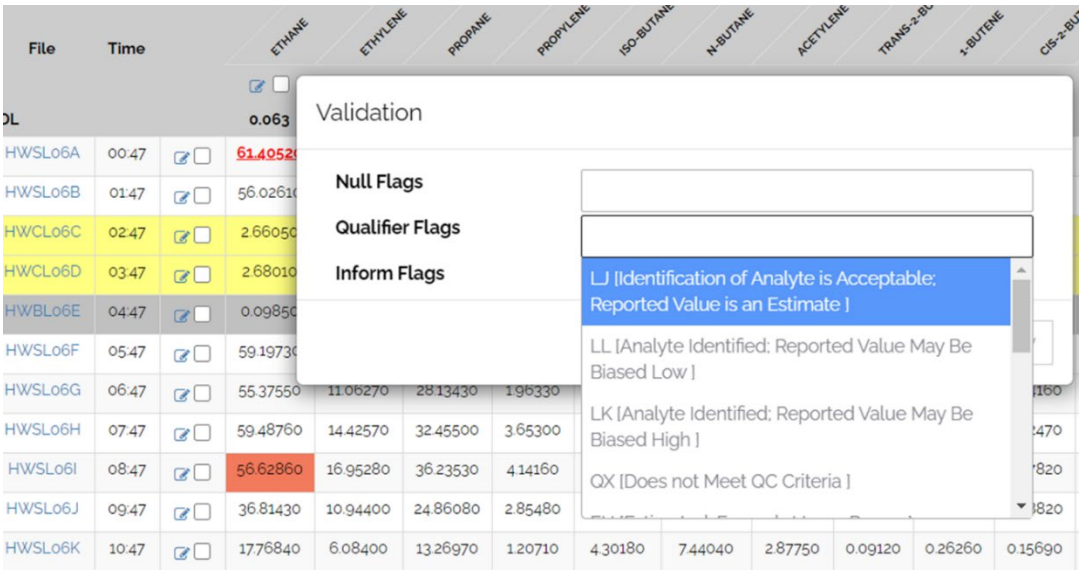


Figure 12. Flagging individual components in MAX

- A whole-hour for a day (whole raw) can be flagged by clicking on the desired raw. The selected raw will be highlighted in red and the validation pop-up menu will pull up when right clicking the mouse. Choose the desired code from the dropdown menu and select "Apply" (**Figure 13**)

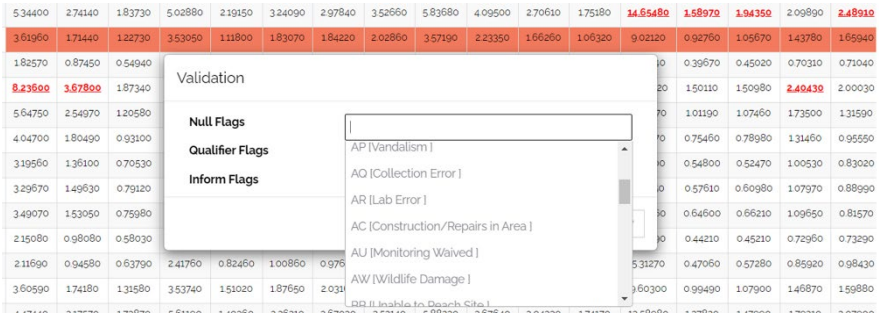


Figure 13. Flagging individual components for entire days in MAX

4.5.3 Cross Time Component Validation and Cross Time Automatic Screening

- The Cross-Time Component Validation option allows the flagged individual or multicomponent by date and hour. Select the Cross-Time Component Validation option and select site, start date & time, component or components and flag options the "Submit" (**Figure 14** and **Figure 15**).

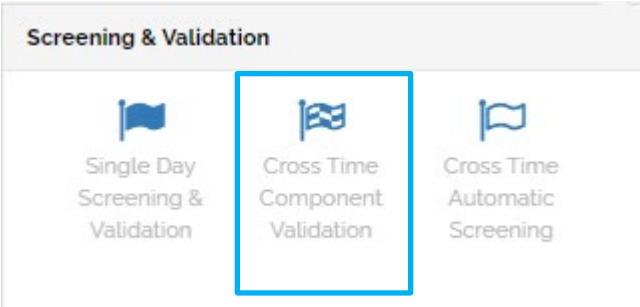


Figure 14 The "Screening & Validation" option in MAX with Cross Time Compound Validation button

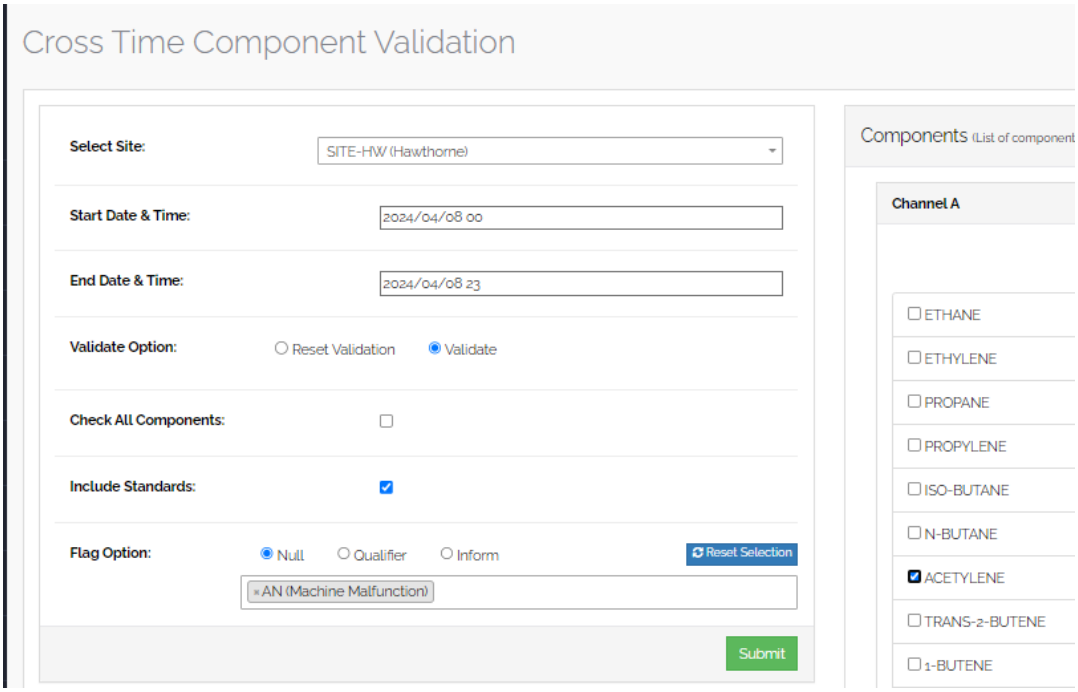


Figure 15. The "Screening & Validation" option in MAX with Cross Time Compound Validation button

- The Cross-Time Automatic Screening option allows to flag individual or multicomponent by date and hour for multiple qualifiers and Null flags. Select the Cross-Time Automatic Screening option and select site, start date & time, component or components and flag options the "Go" (**Figure 16** and **Figure 17**). **Note:** Limit Automatic Screening to one week at a time, and only Qualifier flags or Null flags not both.

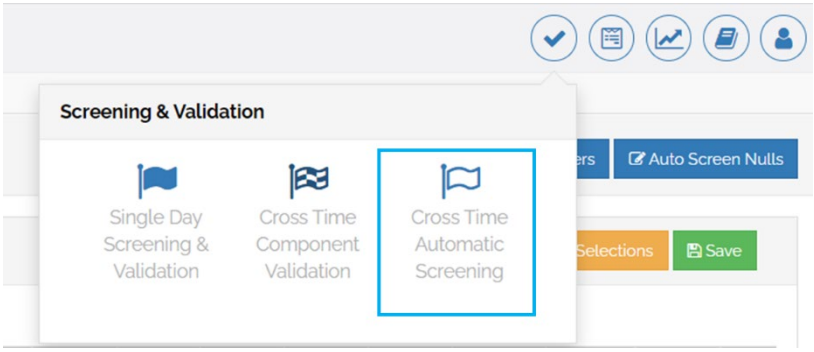


Figure 16. The "Cross Time Automatic Screening" option in MAX

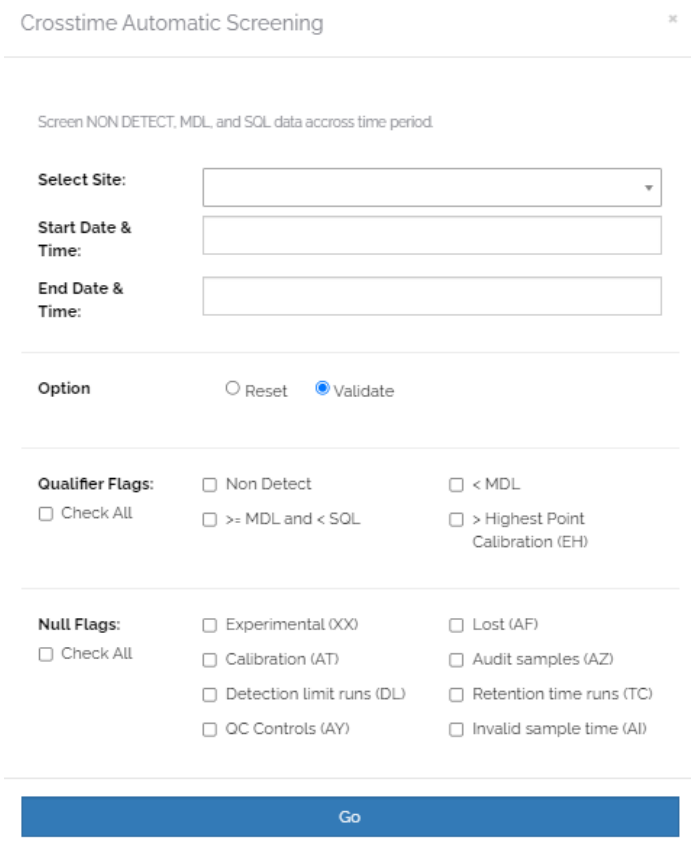


Figure 17. The "Cross Time Screening form in MAX

4.5.4 Invalidating Individual targets using Ambient Time Series Graphs

- The ambient data can also be flagged using Ambient Time Series graphs. Choose the chart report option (**Figure 18**), click on Ambient Air series graph, select the site, start/end date, select either retention times or amounts and components of interest.

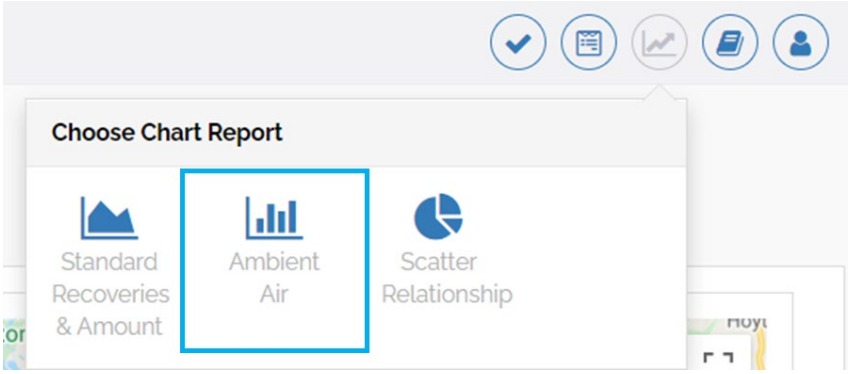


Figure 18. The "Ambient Air Chart Report" option in MAX

- Individual data points may be invalidated by clicking on a data point (which turns the point green), select the "Validate" box, choose the desired code and "Apply". Use the validate button to apply flags when selections are complete.
- The ambient data can be also flagged using the Ambient Time Series graphs option. Select the chart report option (**Figure 19**), click on Ambient Air series graph then select the site, start/end date and either retention times or amounts and components of interest.

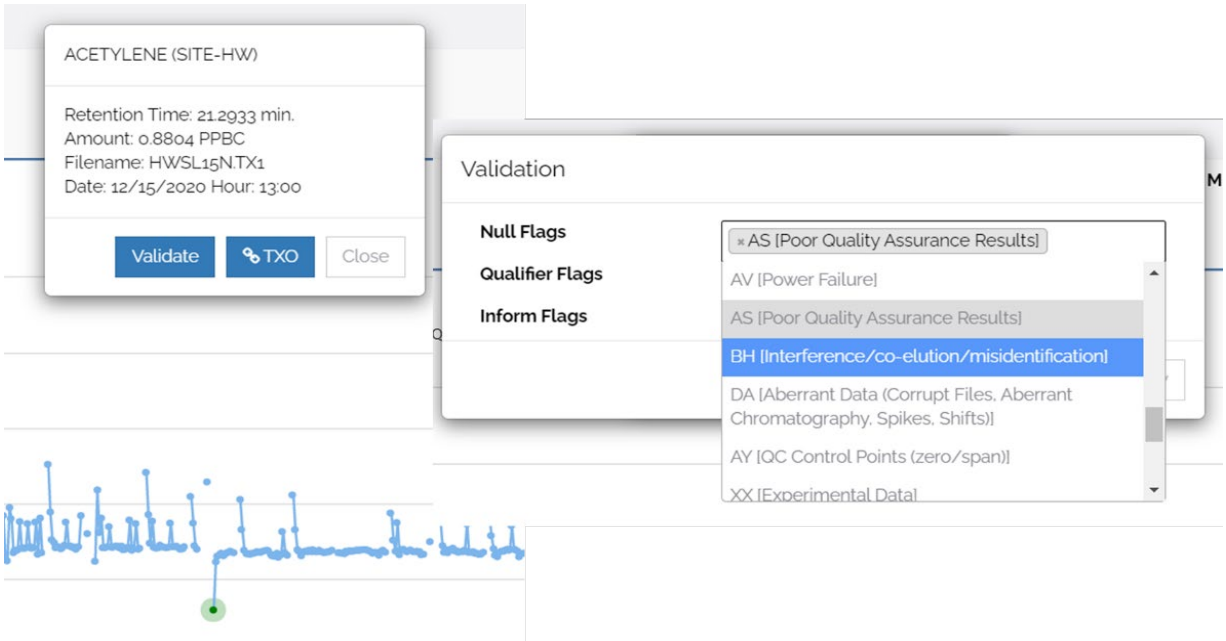


Figure 19. Invalidating individual data points in the time series graph in MAX

4.6 Final Data Quality Review

- The ambient data is reviewed to assure general trends identified are in accordance with the PAMS QAPP and common source tracers for the site.

4.6.1 Locking Data in ORSAT MAX

After the verification, validation, and uploading of reprocessed data is completed for a site-month of data, the data must be locked in ORSAT MAX prior to performing any post-validation procedures.

- Navigate to the relevant site page in ORSAT MAX.
- Scroll down and view the "Monthly Data" tab.
- Select the relevant month of data and click "Go".
- Check the first checkbox in the first column of the Monthly Data tab to select all days in that month.
- Click "Lock" (Figure 20).
    - All days for that month should be outlined in red after locking.
    - Once data is locked, no further changes can be committed to the data.
    - To Unlock and make changes, uncheck the boxes next to the relevant dates and click "Lock". The red outline should be removed.

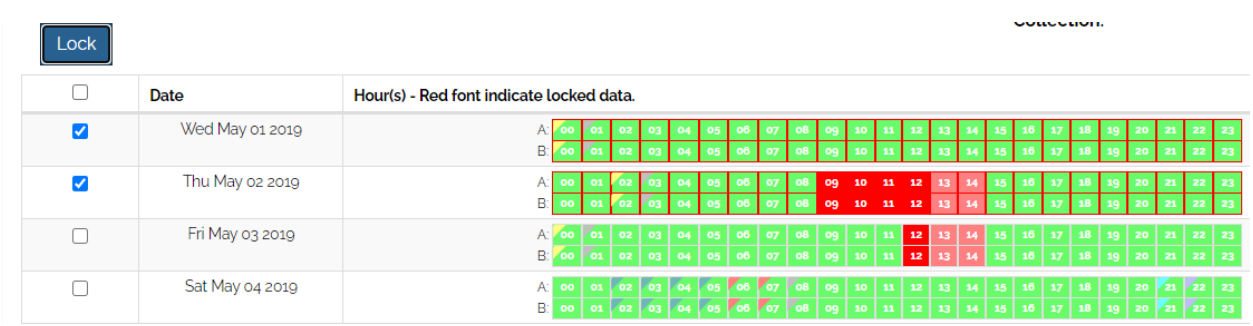


Figure 20. Locking the data in MAX

4.7 Reports

4.7.1 Data Completeness Reports

The data completeness report can be run prior to or after data locking but it needs to be done prior to generating the AQS file because it is a good tool to find errors or flagging that should affect the AQS file size. If any errors are identified in the data files or data flagging, it can be corrected by reloading tx0/tx1 files and reflagging the data after it is reloaded.

- Navigate to the relevant site page in ORSAT MAX.
- Click in the upper right-hand menu bar, select “reports” and “Data Completeness”
- Select the site, start and end dates and click “Go” (**Figure 21** and **Figure 22**)

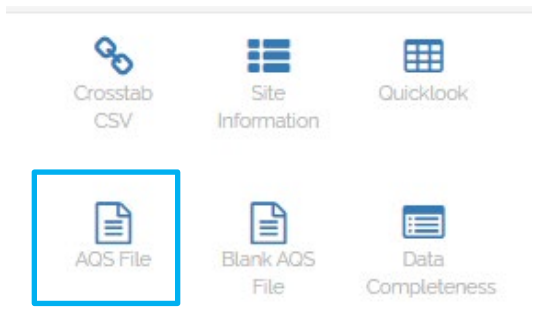


Figure 21. Data Completeness report in MAX

Total Hours In Time Period:	744
Total Collected Hours (Less LST)	726
Runtime Data Recovery (%)	97.58%
Total Null Flagged QC	79
Total Invalid Hours	101
Total Valid Ambient Hours	643
Valid Ambient Recovery (%)	86.42%
Total Valid Ambient Plus QC	722
Valid Ambient Recovery (adj for QC) (%)	97.04%

Figure 22. Data Completeness report and Flagging summary portion in MAX

- Compare the Data Completeness Report to the Validation Notes to ensure all flagging was done correctly

4.7.2 AQS File Generation

The AQS file generation should only be created after the data is locked in MAX

- To generate an AQS file to upload into the EPA database, navigate in the ORSAT MAX page
- Go to the upper right-hand menu and select "Reports"
- Click on AQS file option (**Figure 23**)

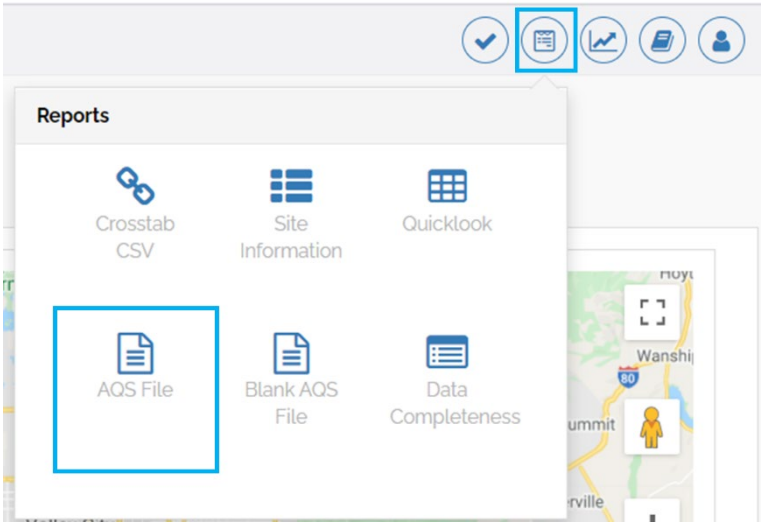


Figure 23. AQS file Report option for Ambient Data

- Select the site, start/stop dates
- Select "Insert", "include QA/QC", "include X Run" and click 'Go"
    - Note that the delete function is used when data has previously been uploaded into the AQS database and it needs to be deleted out of the database (**Figure 24**).

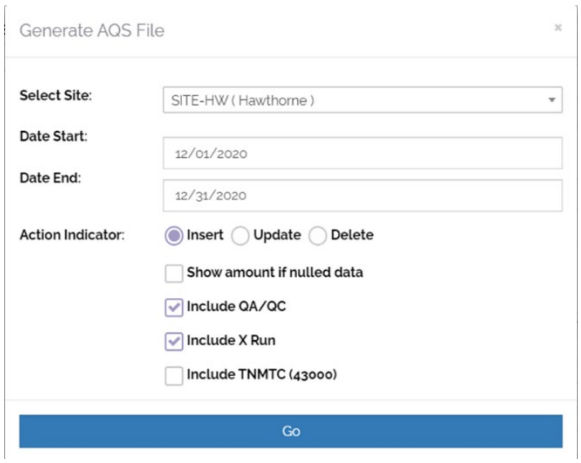


Figure 24. Generate AIRS file form in MAX for Ambient Data

- To generate a Blank AQS File to upload into the EPA database, navigate in the ORSAT MAX page
- Go to the upper right-hand menu and select "Reports"
- Click on Blank AQS file option (see **Figure 25** and **Figure 26**)

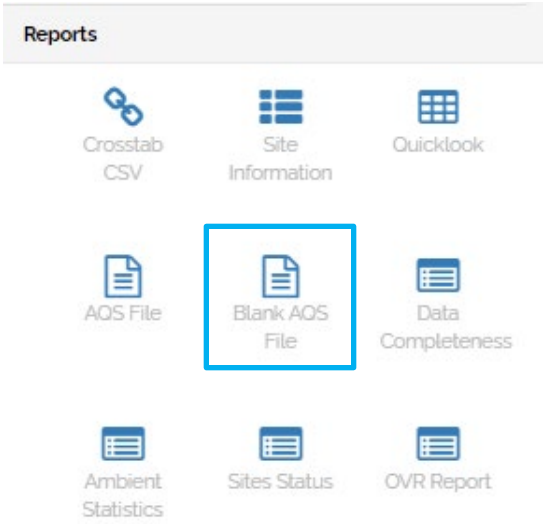


Figure 25. AQS file Report option for Blanks

- Select the site, start/stop dates
- Select "Insert"

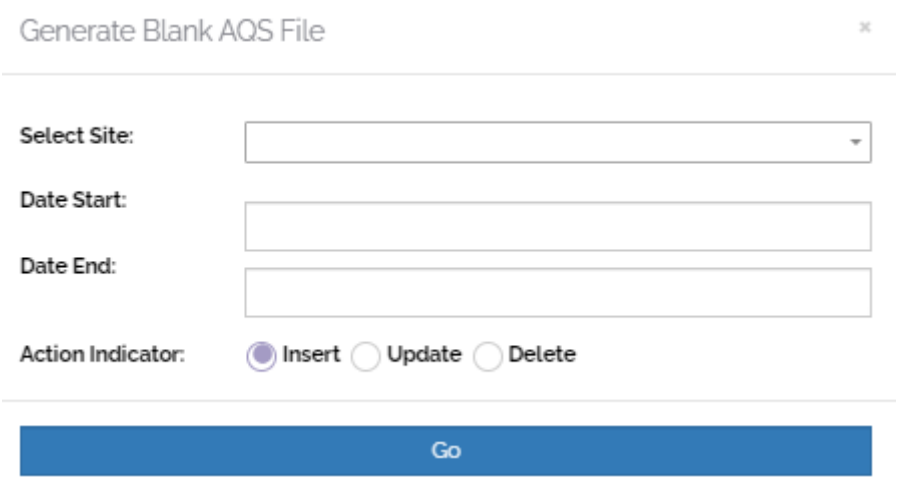


Figure 26. Generate AIRS file form in MAX for Blanks

- Download the CVS file that is generated. The number of lines in the generated report equals the number of hours multiplied by the number target components

4.8 Archiving the Final Data

After a month's data is validated, archive the final monthly data files. Using 7-zip or another ZIP program of choice, compress all *.dat, *.rst, reprocessed *.tx1, validator created *.met, and validator created *.seq files for a given site-month and store them in the U:\PLAN\AMC\GC-Agilent+Markes\Data\FINAL*YYYY* directory. Use the filename format *SiteIDYYYYMMvX*.zip. Record the date archived and the archived file name on the MDVR.

4.9 Calculating Data Recovery

Percent data recovery is a valuable overview of data completeness for a site-month. It is based on the number of valid ambient hours of data out of the total number of hours available in a time period. Because percent data recovery is based on the amount of valid ambient hours, it is a good indicator of the status of an instrument even though it is not a measure of how often individual compounds are invalid during a time period

The percent data recovery can be calculated with the following equation:

$$\%_{Data\,Recovery} = \frac{(24\,Hours \times Days\,in\,Month) - (Number\,of\,Invalid\,Hours)}{(24\,Hours \times Days\,in\,Month)} \times 100\%$$

Percent data recovery may also be calculated for individual compounds if necessary or requested, but it is not required.

Document the percent data recovery on the MDVR. Non-ambient QC hours are accounted for in the total percent data recovery when added to the MDVR.

4.10 Finalizing the Monthly Data Validation Report

Review the MDVR for a given site-month and ensure that all information has been completed and is accurate. The MDVR is then printed to pdf to lock the data on the sheet. If corrections must be made to a month of data following initial validation, note all corrections and make a note of the reason re-validation was required. Make a PDF file of the completed MDVR and save it to U:\PLAN\AMC\GC-Agilent+Markes\Data\MDVR*YYYY*. After transferring the MDVR to the U:\, email the operator and managers to let them know that it is completed.

4.11 Maintaining Network Reference Tables in ORSAT MAX

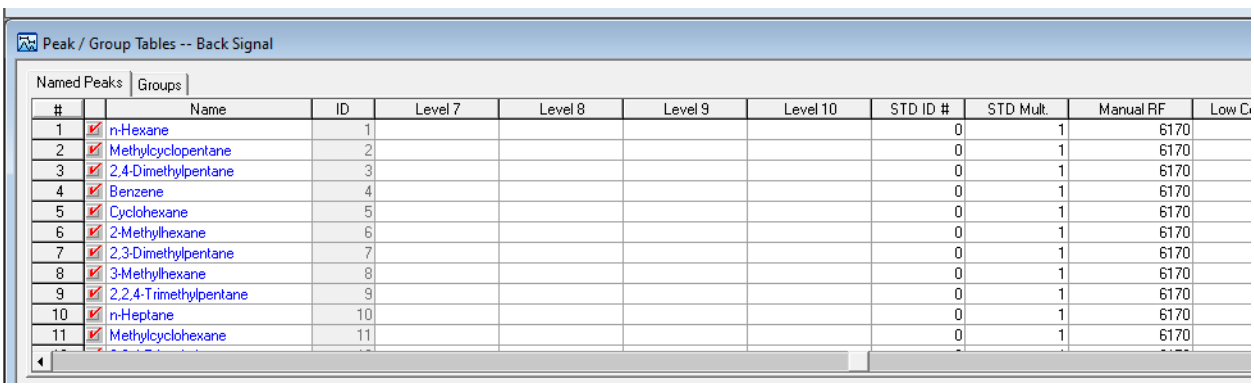
There are multiple network reference tables in ORSAT MAX that will occasionally need to be adjusted. Care should be used when making any changes to these tables; only experienced AutoGC data validators or the subject matter expert should perform these tasks. Any change to these tables should also be reflected in an email to the site operator, subject matter expert, quality assurance coordinator, TAS section manager, and AMS section manager. Tables that occasionally need to be adjusted include the CVS Standards, LCS Standards, RTS Standards, MDL Standards, Site Information, Network Target Components, and Network Standard Components.

5.0 CALCULATIONS

5.1 Compound Concentrations

Compound concentrations are calculated in EZChrom automatically by dividing peak area counts by response factors that are manually entered after calibration by the site operator as part of the instrument method. This response factor needs to be updated in any methods from the date of calibration forward.

- There are two places in the method that needs to be updated to properly calculate the values based on the response factors.
  1. In the Peaks/Groups table of the method in the Named Peaks tab. It is in the Manual RF column that is about halfway through the list of columns (**Figure 27**).



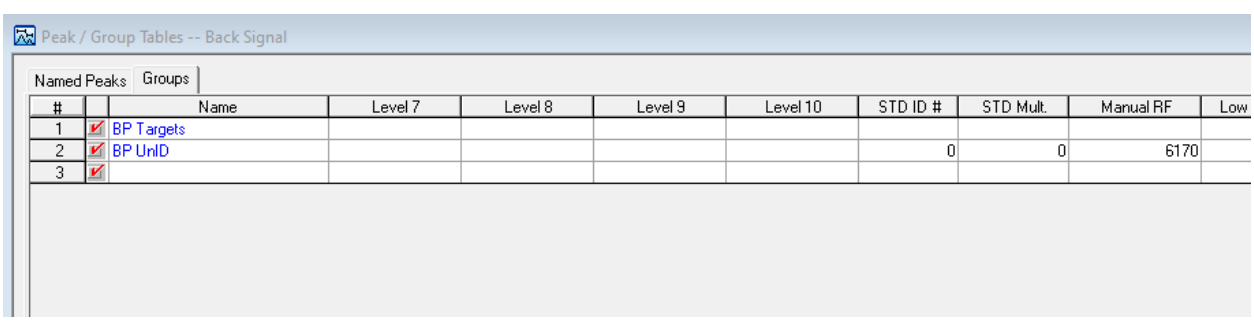
Peak / Group Tables -- Back Signal

Named Peaks | Groups

#		Name	ID	Level 7	Level 8	Level 9	Level 10	STD ID #	STD Mult.	Manual RF	Low C
1	✓	n-Hexane	1					0	1	6170	
2	✓	Methylcyclopentane	2					0	1	6170	
3	✓	2,4-Dimethylpentane	3					0	1	6170	
4	✓	Benzene	4					0	1	6170	
5	✓	Cyclohexane	5					0	1	6170	
6	✓	2-Methylhexane	6					0	1	6170	
7	✓	2,3-Dimethylpentane	7					0	1	6170	
8	✓	3-Methylhexane	8					0	1	6170	
9	✓	2,2,4-Trimethylpentane	9					0	1	6170	
10	✓	n-Heptane	10					0	1	6170	
11	✓	Methylcyclohexane	11					0	1	6170	

Figure 27. Manual RF location in the Name Peak Tab.

  2. In the Peaks/Groups table of the method in the Groups tab. It is in the Manual RF column that is about halfway through the list of columns (**Figure 28**).



Peak / Group Tables -- Back Signal

Named Peaks | Groups

#		Name	Level 7	Level 8	Level 9	Level 10	STD ID #	STD Mult.	Manual RF	Low
1	✓	BP Targets								
2	✓	BP UnID					0	0	6170	
3	✓									

Figure 28. Manual RF location in the Group Tab.

5.2 Percent Difference

Accuracy is the degree to which the result of a measurement, calculation, or specification conforms to a reference value, therefore reflecting elements of both bias and precision. A term routinely used to describe accuracy is the percent difference between the theoretical value (X_i) and the analyzer response (Y_i) for each instance (i) that the analyzer is challenged.

For the PAMS AutoGC program, accuracy is determined by challenging the AutoGC with gas standards containing selected components of interest at typical, representative ambient concentrations. The gas standard component concentrations are within five to ten times the estimated detection limits. The control standard used for the PAMS AutoGC program is the 15-component gas standard known as the CVS.

Data for accuracy is derived from analysis of the CVS on a daily basis. The "theoretical value" is termed the expected value since it is the value expected for the concentration of each component for a given instance. The "analyzer response" is termed the measured value since it is the value measured by the AutoGC for the concentration of each component for the given instance. The calculation for the percent difference is:

$$Percent\ Difference = \frac{X_i - Y_i}{X_i} * 100$$

where:

i = *a given instance, e.g., the time of measurement*

X_i = *the expected (theoretical) value*

Y_i = *the measured (analyzer response) value*

For practical purposes, the PAMS AutoGC program uses percent recovery as the accepted measure of accuracy, which may be calculated as:

$$Percent\ Recovery = \frac{Y_i}{X_i} * 100$$

where:

i = *a given instance, e.g., the time of measurement*

X_i = *the expected (theoretical) value*

Y_i = *the measured (analyzer response) value*

For example:

$$Percent\ Recovery = \frac{36.69\ ppbC}{33.34\ ppbC} * 100 = 110.05\%$$

5.3 Precision

Precision is a measure of agreement among two or more determinations of the same parameter under similar conditions and can be described as the repeatability of results. Estimates of precision are assessed in different ways for different measurement technologies. For the PAMS AutoGC program, data for precision is derived from duplicate analysis of the daily CVS on a weekly basis. The term routinely used to describe precision is relative percent difference (RPD), which is based on the values derived from back-to-back CVS runs. The calculation for precision using duplicate measurements is:

$$RPD = \frac{X_1 - X_2}{(X_1 + X_2)/2} * 100$$

where:

RPD = relative percent difference

X_1 = the larger of the two values

X_2 = the smaller of the two values

For example**:**

$$RPD = \frac{36.69\ ppbC - 35.98\ ppbC}{(36.69\ ppbC + 35.98\ ppbC)/2} * 100 = 1.95\%$$

5.4 Percent Data Recovery

Percent Data Recovery is the measure of data completeness utilized by the Utah DAQ AutoGC program.

$$\%_{Data\ Recovery} = \frac{(24\ Hours \times Days\ in\ Month) - (Number\ of\ Invalid\ Hours)}{(24\ Hours \times Days\ in\ Month)} \times 100\%$$

5.5 Method Detection Limits (MDLs)

MDLs are needed to provide method sensitivity information to the data user. It is calculated annually prior and after each PAMS sampling season. It is estimated by analyzing a series of low concentration standard sample replicates (minimum 7) and a series of blanks (minimum 7). The spiked samples and blanks are prepared and analyzed over the course of three or more different dates and an average and standard deviation are calculated from the resulting spike data and blank data. The calculated spike standard deviation is multiplied by an appropriate Student's T statistic according to the number of spiked samples to calculate an MDL for the spikes, MDL*sp*. The blank standard deviation is multiplied by an appropriate Student's T statistic according to the number of blanks and added to the average blank value to calculate an MDL for the blanks, MD*Lb*. The higher of the MDL*sp* and MD*Lb* is reported for the site MDL for the target analyte.

Table 9. One-sided 99th Percentile Student's t Values

Number of MDL Samples (n)	Degrees of Freedom ν (n-1)	Student's t Value
7	6	3.143
8	7	2.998
9	8	2.896
10	9	2.821
11	10	2.764
12	11	2.718
13	12	2.681
14	13	2.650
15	14	2.624
16	15	2.602
17	16	2.583
18	17	2.567
19	18	2.552
20	19	2.539
21	20	2.528
22	21	2.518
23	22	2.508
24	23	2.500
25	24	2.485
26	25	2.479
27	26	2.473
28	27	2.467
29	28	2.462
30	29	2.457
31	30	2.453
32	31	2.449
33	32	2.445
34	33	2.441

Note: Acetylene are missing in all of the spikes and the pinenes (alpha and beta) are not sufficiently present or missing in all the spike runs to calculate their standard deviations then the MDL. To estimate the MDLs for these three compounds it is possible to either use the 0.5 default value suggested by the PAMS TAD or use the average MDL value for the corresponding column from which each compound is eluting. The averaged concentrations for each of the species (acetylene, alpha, and beta pinene) are most of the time above the averaged MDL for each of the columns, so using the averaged MDL values for the PLOT column as MDL value for the Acetylene and the averaged MDL values for the BP column as MDL value for alpha and beta pinene will be a reasonable approach.

6.0 DEFINITIONS

6.1 Acronyms

AQS	Air Quality System (EPA database)
AMCV	Air Monitoring Comparison Values
AMS	Air Monitoring Section
AutoGC	Automated Gas Chromatograph
CDC	Continuing Demonstration of Capability
CVS	Calibration Verification Standard
DAQ	Division of Air Quality
DEQ	Department of Environmental Quality
MDL	Method Detection Limit
EPA	Environmental Protection Agency
IDC	Initial Demonstration of Capability
LCS	Laboratory Control Standard
MDVR	Monthly Data Validation Report
MAX	Merlin AutoGC Xplorer
PAMS	Photochemical Assessment Monitoring Station
ppb	Part Per Billion
ppbC	Part Per Billion Carbon
ppbV	Part Per Billion Volume
QA	Quality Assurance
QAPP	Quality Assurance Project Plan
QC	Quality Control
RF	Response Factor
RPD	Relative Percent Difference
RTS	Retention Time Standard
SOP	Standard Operating Procedure
VOC	Volatile Organic Compound

6.2 Interchangeable Terms

For the purpose of the DAQ AutoGC program, the following terms are interchangeable:

- TNMHC and TNMOC – For purposes of the DAQ AutoGC data, these terms are used interchangeably and represent the sum total of the non-methane hydrocarbon compounds (EPA parameter code 43102). The TNMHC is equivalent to the TNMTC plus unidentified compound concentrations.
- CVS and 1-ppmV Standard – The CVS, performed daily, is sometimes referred to as the "1-ppmV Standard", "Primary Standard", "Verification Standard", or "Daily Calibration" by various program participants.
- LCS and 5-ppbV Standard – A second source "Laboratory Calibration Standard (LCS)" is performed weekly at the sites and is often referred to as the "5-ppbV Standard" or "Second Source Standard" by various program participants.
- RTS, RTVS, and 57-Component Standard – A "Retention Time Standard (RTS)" is performed weekly and is also known as the 57-Component Standard. In addition, some program participants refer to this standard as the "Retention Time Verification Standard (RTVS)."
- Column A, Channel A, Front Channel, Front Signal, PLOT, PL, and Porous Layer Open Tubular Column – The "lightest" VOCs (methane to 1-hexene) will elute out on the "PLOT column," which may also be referred to as "Column A" or "Channel A."
- Column B, Channel B, Back Channel, Back Signal, BP, and Boiling Point Column – Starting with n-hexane, the "heavier" VOCs will elute out on the "Boiling Point column," which is often referred to as "BP" or "Column B."

6.3 Equivalencies

For the purpose of the DAQ AutoGC program, the following equivalencies are in use:

- TNMHC equals TNMTC plus all unidentified components (UNID).
- When TNMHC is invalid, this is equivalent to both TNMTC and all unidentified components (UNID) being invalid. For instance, when

an entire column is flagged invalid, the target compounds, UNID components, TNMHC, and TNMTC are also flagged as invalid.

7.0 REFERENCES

Photochemical Assessment Monitoring Station (PAMS) Network Quality Assurance Project Plan (QAPP) for Utah Air Monitoring

Texas Commission on Environmental Quality, Monitoring Division SOP DV-002-VOC, Revision 0, Automated Gas Chromatograph (AutoGC) Data Collection, Management, Validation, and Reporting, July 2018

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U.S.EPA. Technical Assistance Document for Sampling and Analysis of Ozone Precursors for the photochemical Assessment Monitoring Stations Program. April 2019. Available at:

https://www.epa.gov/sites/production/files/2019-11/documents/pams_technical_assistance_document_revision_2_april_2019.pdf

U.S EPA. Quality Assurance Project Plan for the Photochemical Assessment Monitoring Stations (PAMS) Required Site Network for Speciated Volatile, Organic Compounds, Carbonyls, and Meteorological Parameters Including Mixing Layer Height-April 2019

https://www.epa.gov/sites/production/files/2021-03/documents/ard_pams_model_qapp_2020-signed.pdf

8.0 REVISION HISTORY

Draft Revision 1. Effective Date: April 01, 2023

Draft Revision 2. Effective Date: May 01. 2024

Revision 3. Effective January 2025

APPENDIX A. UTAH DAQ MASTER AUTOGC COMPOUND LIST

Priority Compounds		Optional Compounds	
1,2,3-trimethylbenzene [a]	m-ethyltoluene [a]	1,3,5-trimethylbenzene	isopropylbenzene [b]
1,2,4-trimethylbenzene [a]	n-butane	1-pentene	m-diethlybenzene
1-butene	n-hexane [b]	2,2-dimethylbutane	methylcyclohexane
2,2,4-trimethylpentane [b]	n-pentane	2,3,4-trimethylpentane	methylcyclopentane
benzene [a,b]	o-ethyltoluene [a]	2,3-dimethylbutane	n-decane
cis-2-butene	o-xylene [a,b]	2,3-dimethylpentane	n-heptane
ethane [c]	p-ethyltoluene	2,4-dimethylpentane	n-nonane
ethylbenzene [a,b]	propane	2-methylheptane	n-octane
ethylene	propylene	2-methylhexane	n-propylbenzene [a]
isobutane	styrene [a,b]	2-methylpentane	n-undecane
isopentane	toluene [a,b]	3-methylheptane	p-diethylbenzene
isoprene	trans-2-butene	3-methylhexane	trans-2-pentene
m&p-xylenes [a,b]	total non-methane organic carbon (TNMOC)	3-methylpentane	**α/β-pinene**
		acetylene	1,3 butadiene [b]
		cis-2-pentene	cyclohexane
		cyclopentane	

Source: Revisions to the Photochemical Assessment Monitoring Stations Compound Target List. U.S. EPA, November 20, 2013

[a] Important SOAP (Secondary Organic Aerosols Precursor) Compounds

[b] HAP (Hazardous Air Pollutant) Compounds. [c] Non-reactive compounds, not considered to be VOC for regulatory purposes

APPENDIX B. LIST OF EPA/AQS NULL DATA AND QUALIFIER CODES

	CODE	Description	Example	Range
NULL CODES	DA	Aberrant Data (Corrupt Files, Aberrant Chromatography, Spikes, Shifts)	Run collected but chromatography incomplete or not integrated	Invalid whole hours
	BK	Site computer/data logger down	optional LST data code	
	AN	Machine malfunction	Same as DA? May have values or may not	
	AF	sample scheduled but not collected	LST data (Auto Flag) system not running	
	XX	Experimental Data	Test data or bakeout; X runs (Auto Flag)	
	BJ	Operator Error	alternate code for XX	
	BA	Routine maintenance	alternate code for XX	
	AQ	Collection error	Failed QC run- looks like ambient or first run after system restarts or power failure, other Markes log issue	
	AG	Sample time out of limits	Sampling time on Markes log indicates < 40 minutes	
	AI	Insufficient Data (cannot calculate)	Sample collection for less than 75% of sample hour. Auto Flag	
	AH	Sample Flow Rate out of Limits	Markes log	
	AM	Miscellaneous Void		
	AT	Calibration	M runs, calibration curves	QC Samples (Auto Flag)
	AX	Precision Check	Applies when data represent instrument precision check	
	AY	QC Control Points (zero/span)	SSCV, CCV runs and Blank runs; Applies when data represent instrument QC checks	
	AZ	QC Audit	T runs; PE samples	
	DL	Detection Limit Analyses	D runs; MDL samples	
	TC	Component Check & Retention Time Standard	Q runs; Retention Time Standards runs	
	AS	Poor QA results	Totals > 10 ppbC; targets below failing recovery in QC for 2 tier validation criteria	Invalid targets
	BH	Interference/co-elution/misidentification	invalid target due to miss ID or poor integration, isolated spike etc.	

	CODE	Description	Example	Range
Qualifiers	*5*	Outlier	Individual outlier data - may choose to invalidate	Individual Targets
	QX	Does not meet QC criteria	Acetylene < 55% recovery	
	LL	Analyte Identified; Reported Value May Be Biased Low	targets < 70% in RTS	
	LK	Analyte Identified; Reported Value May Be Biased High	m-etol - interference/coelution; targets > 120% recovery in RTS	
	LJ	Identification of Analyte is Acceptable; Reported Value is an Estimate	poor integration or same as LK/LL	
	LB	Lab blank value above acceptable limit	flag targets failing in blank back and forward	
	EH	Estimated; Exceeds Upper Range	> highest curve value	
	VB	Value below normal; no reason to invalidate	?? May consider null based on sample volume log or other consideration	
	V	Validated value	Data should be validated when uploaded to AQS, this code is not necessary but may identify suspect values that have gone through additional scrutiny	
	MD	Value less than MDL	Less than MDL	Auto Flag
	ND	No Value Detected	all zero data	
	SQ	Values Between SQL and MDL	Values between SQL and MDL	